SkelGym: Cross-Paradigm Kinematic Fusion and Dynamic Augmentation for 22-Class Gym Exercise Classification

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Abstract

Automated gym exercise recognition is critical for developing intelligent virtual personal trainer systems capable of monitoring workout execution, tracking training volume, and mitigating musculoskeletal injury risks. However, traditional RGB video approaches incur prohibitive computational costs, suffer from environmental background overfitting, and raise severe privacy concerns when streaming from home environments. We evaluate on a curated corpus of 1,024 in-the-wild videos spanning 22 fine-grained resistance exercises under strict video-level partitioning with no source-video overlap across train, validation, and test sets. We present **SkelGym**, a lightweight, privacy-aware intelligent exercise recognition framework operating exclusively on 3D skeletal coordinates extracted via monocular pose estimation. SkelGym introduces four primary technical contributions: (1) a curated, multi-source 3D skeletal benchmark spanning 1,024 videos across 22 resistance exercises under strict video-level partitioning with zero split leakage; (2) a 117-dimensional *Biomechanical Compound Representation* fusing relative 3D joint displacements with pairwise elevation and azimuth angles; (3) *SkelGym-Aug*, an empirically selected, task-oriented augmentation configuration specifically tailored for resistance exercise biomechanics through controlled Leave-One-Out ablation; and (4) a cross-paradigm ensemble coupling a self-attention Transformer with a four-stream Adaptive Graph Convolutional Network via

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validation-calibrated SLSQP soft voting, achieving real-time sub-millisecond edge classification. SkelGym achieves **69.74% ± 1.04% window-level accuracy** and **79.11% ± 0.25% video-level consensus accuracy** (Video Macro F1: **0.7834 ± 0.0082**, Window Macro F1: 0.6882 ± 0.0068) across three independent random seeds under fixed splits ($N = 2,743$ test windows from 233 held-out videos), with selected model comparisons evaluated via paired statistical tests (McNemar $p < 10^{-12}$, Wilcoxon $p < 0.005$). With an ultra-low classifier inference latency of merely **0.42–4.33 ms** on host CPU (0.08–0.54 ms on CUDA, 1.11–8.77 ms on MPS; where single-model backbones execute in 0.42–1.16 ms and the full five-stream ensemble executes in 4.33 ms) per 32-frame window, SkelGym enables real-time, privacy-aware exercise recognition suitable for smart mirrors, mobile devices, and embedded gym displays. All source code, extracted skeletal representations, metadata, and pretrained checkpoints used for reproducibility are publicly available at `https://huggingface.co/Cuong2004/gym-exercise-classification` co the phai chuyen link sang github.

Keywords: Gym exercise classification, Skeleton-based action recognition, Adaptive Graph Convolutional Networks, Skeletal Transformers, Biomechanical features, Real-time inference, Edge AI

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1. Introduction

Regular physical resistance training and structured exercise play an essential role in improving cardiovascular health, metabolic longevity, muscular hypertrophy, and psychological well-being [39]. With modern demanding lifestyles and remote working environments, people prefer conducting workouts in home rather than going to commercial fitness centers. However, doing gym exercise without the supervision of certified coaches frequently leads to improper kinetic execution, sub-optimal muscular recruitment, and elevated risks of acute or repetitive strain injuries [34]. Consequently, there is an urgent demand for automated, intelligent virtual personal trainers capable of identifying exercise types, monitoring execution repetitions, and providing instantaneous biomechanical feedback [49].

Early computer vision frameworks for resistance exercise and action recognition were proposed upon appearance-based RGB video models, such as 2D/3D CNNs [1, 2, 3] and spatiotemporal Vision Transformers, e.g., ViViT [4], TimeSformer [5], Video Swin [6]. While dense RGB representations cap-

ture visual cues such as gym equipment and barbells [7], they exhibit three critical operational drawbacks in practical deployments. First, dense spatio-temporal convolutions impose heavy computational overhead, preventing real-time execution on edge devices. Second, appearance-based networks are dominated by background cues, causing performance degradation in unseen environment. Third, cloud-centric video architectures require continuous high-definition video transmission streams, introducing network bandwidth bottlenecks and privacy risks.

To address these challenges, skeleton-based human action recognition offers an elegant solution by abstracting human bodily movement into sparse, lightweight 3D keypoint coordinates. With real-time pose estimation frameworks such as Google MediaPipe [8, 9], OpenPose [10], and AlphaPose [11], 3D skeletal tracking can be executed on edge devices at sub-millisecond latencies. However, translating skeletal tracking into accurate, multi-class gym exercise recognition introduces three fundamental scientific challenges. The first is balancing biomechanical redundancy against kinematic expressiveness. Distinguishing fine-grained exercise variants requires capturing subtle joint-angle dynamics. In addition, exhaustive geometric formulations—such as computing all three-joint triplets can lead to a combinatorial feature explosion, causing high multicollinearity, heavy computational overhead, and severe overfitting risks. The second is from physiologically invalid data augmentation. Standard spatial transformations distort joint limits and fixed bone lengths, producing anatomically impossible poses that corrupt feature learning. Finally, the third is achieving true cross-paradigm model complementarity while preventing data contamination. Temporal architectures, such as LSTM [12, 13] and transformer [14], model long-range phase transitions. Whereas spatio-temporal graph convolutional networks, such as ST-GCN [16], 2S-AGCN [17], and CTR-GCN [20], exploit physical musculoskeletal topology. Naive combination or tuning ensemble fusion weights on test sets introduces optimistic reporting bias.

To systematically resolve these challenges, SkelGym, a scientifically grounded cross-paradigm skeletal classification framework, is proposed. The contributions of this work are as follows:

- A compact 117-dimensional kinematic descriptor is proposed for classifying gym exercises. Specifically, the proposed descriptor unifies relative 3D joint coordinates (39D) with pairwise elevation angles (78D). This biomechanical compound representation simultaneously captures

Euclidean spatial metrics and scale-invariant directional orientation.

- To preserve biomechanics and gravity constraints, a task-aware data augmentation method combining sagittal reflection, angular transformations, and 3D yaw rotations is applied. While temporal warping distorts execution cadence and degrades recognition, geometric transformations and light sensor noise effectively build robustness against variations in body size, camera angle, and sensor jitter. Consequently, we isolate an optimal 4-operator configuration that strictly enforces musculoskeletal and gravitational constraints during training, stabilizing model performance and significantly dampening optimization variance. –> khoong hieu lam

- We introduce a 3D Skeletal landmark benchmark derived from 1,024 multi-camera in-the-wild videos spanning 22 resistance training exercises. The dataset aggregates three distinct collection sources: public repositories, licensed stock media, and 128 ethically collected author-recorded gym recordings. Furthermore, to address temporal frame-level data leakage common in existing fitness benchmarks, we enforce a strict video-level partitioning protocol (6:2:2 train/val/test split) ensuring zero cross-split source video overlap.

- To achieve real-time, privacy-preserving activity recognition, we introduce a cross-paradigm ensemble unifying lightweight self-attention Transformers and a four-stream Adaptive Graph Convolutional Network (AAGCN). Multi-stream prediction probabilities are integrated through a validation-calibrated, leakage-free SLSQP soft-voting optimization [27, 28]. The framework is evaluated across 22 resistance exercises with paired statistical validation. SkelGym attains 69.74% $\pm$ 1.04% window-level accuracy and 79.11% $\pm$ 0.25% video-level accuracy (Video Macro F1: 0.7834 $\pm$ 0.0082). Crucially, individual backbones execute in 0.42–1.16 ms on CPU, while the full five-stream ensemble incurs a total classifier latency of just 4.33 ms on CPU (0.54 ms on CUDA GPU), highlighting its applicability to edge devices while preserving privacy without continuous video recording.

2. Related Work

2.1. Skeletal Action Recognition

Early landmark-based action recognition utilized Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [12, 13, 15] to model skeletal temporal dynamics. However, pure recurrent models treat skeletons as unorganized coordinate vectors, ignoring the inherent topological structure of the human musculoskeletal system. To overcome this, Yan et al. [16] pioneered Spatial Temporal Graph Convolutional Networks (ST-GCN), constructing a spatio-temporal graph where human joints form vertices and physical bones form fixed spatial edges. Shi et al. [17] introduced Two-Stream Adaptive Graph Convolutional Networks (2S-AGCN), parameterizing the adjacency matrix with learnable and data-driven components. Subsequent advances, including multi-stream AAGCN [18], CTR-GCN [20], InfoGCN [47], HD-GCN [48], and SkateFormer [21], progressively integrated channel attention, information-theoretic representation learning, hierarchical graph decomposition, and partition-specific transformer attention. In another direction, Duan et al. [56] introduced PoseConv3D, mapping 3D skeleton keypoints to spatiotemporal heatmap volumes to exploit 3D convolutional architectures, while Yin et al. [50] designed an efficient multi-stream depthwise separable convolutional network for skeleton action recognition. However, these modern architectures were predominantly tuned around large-scale benchmarks like NTU RGB+D (60/120 classes) with parameter numbers of 1.3M–5.2M and high GFLOP demands.

2.2. Kinematic Sequence Modeling and Skeletal Transformers

Following the success of self-attention in natural language processing (NLP) and computer vision [14, 4, 5, 6], transformers have been adapted for skeletal motion [19]. However, despite global receptive fields, transformers are data-hungry and prone to overfitting on small skeletal datasets.

2.3. Skeletal Data Augmentation in Action Recognition

Data augmentation is essential for mitigating overfitting in deep skeleton-based action recognition, particularly under constrained training regimes [16, 18]. Existing augmentation strategies typically fall into three broad categories: spatial-geometric transformations (random 3D rotations, sagittal flipping, coordinate scaling) [], temporal manipulations (temporal cropping, time warping, speed perturbation) [], and noise injection (additive

Gaussian coordinate jitter) [23]. Although these general transformations enhance model robustness, they often ignore domain-specific biomechanical constraints. Consequently, unconstrained 3D rotations disrupt gravitational alignment, non-uniform spatial scaling distorts invariant anatomical bone proportions, and arbitrary temporal manipulation risks corrupting movement execution dynamics.

2.4. Fitness and Resistance Exercise Recognition

Recently, automated exercise tracking systems have attracted growing attention in sports science, rehabilitation, and ubiquitous computing. García-de-Villa et al. [49] developed an expert system for simultaneous exercise recognition and evaluation in prescribed fitness routines for virtual coaches. Strömbäck et al. [51] created MM-Fit, a multimodal benchmark unifying 3D pose, multi-view RGB-D video, and wearable inertial sensors across free-weight exercises, demonstrating the utility of multi-sensor tracking. Riccio [34] implemented real-time exercise classification and repetition counting using a BiLSTM over geometric joint angles. Ding et al. [36] developed the RepCount benchmark for repetition counting in diverse fitness environments. Deyzel et al. [46] proposed one-shot skeleton-based recognition for strength and conditioning exercises on the SU-EMD dataset, revealing critical confusion between squats and deadlifts due to kinematic overlap. While general benchmarks (UCF101 [37], Kinetics-400 [38]) feature broad athletic activities, they lack fine-grained biomechanical resolution.

3. Dataset and Experimental Protocol

3.1. Data Source, Provenance, and Ethical Compliance

The experimental dataset consists the landmark extracted from 1,024 clean, un-shuffled video recordings across 22 resistance training exercises, including barbell biceps curl, bench press, chest fly machine, deadlift, decline bench press, hammer curl, hip thrust, incline bench press, lat pull-down, lateral raise, leg extension, leg raises, plank, pull-up, push-up, Romanian deadlift, Russian twist, shoulder press, squat, t-bar row, tricep dips, and tricep pushdown. The exercises are shown in Fig. 1. To ensure high diversity across environmental backgrounds, camera perspectives, illumination conditions, and athlete anthropometry, the dataset unifies three distinct provenance streams: (1) public video repositories, comprising 652 videos recovered from the open Workout/Exercise Video Dataset [35]; (2) curated

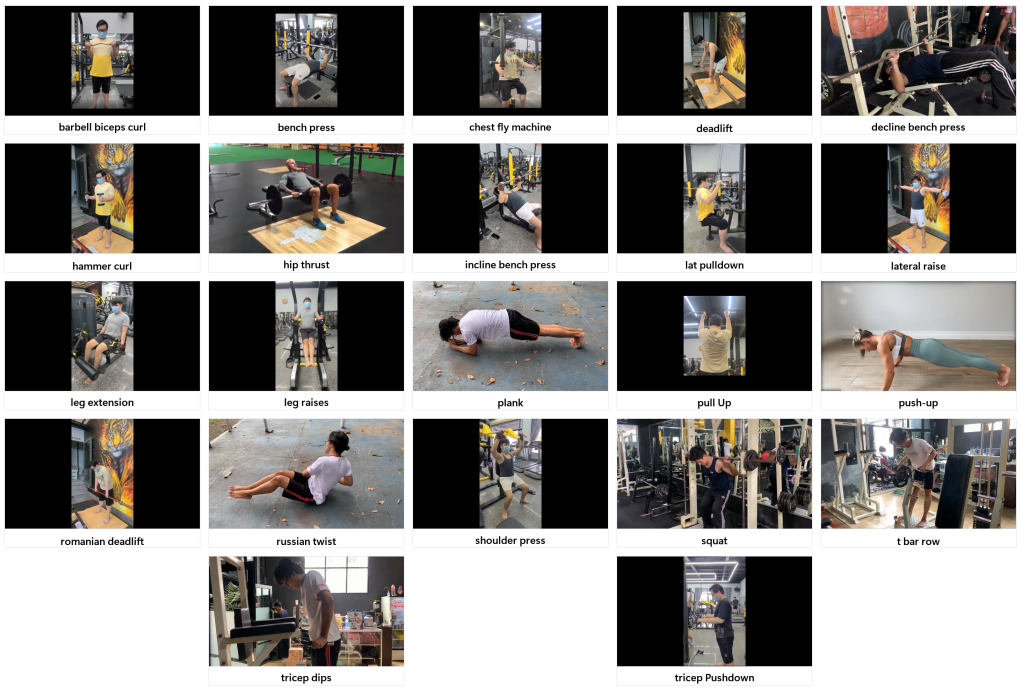


Figure 1: Taxonomy and sample distributions across the 22 resistance exercises in the benchmark dataset.

Table 1: Key specifications, provenance breakdown, and capture characteristics of the GYMED exercise dataset.

Specification / Characteristic	**Value / Distribution**
Total Videos / Total File Volume	1,024 unique video recordings / 10,225.37 MB ($\approx$ 10.2 GB)
File Formats & Video Encoding	MP4 (964 videos, 94.1%), MOV (60 videos, 5.9%) / H.264
Frame Rate & Recording Duration	30 frames per second (FPS) / 20–30 s per recording
Trimmed Action Segments	1,108 clean exercise segments (average 1.09 segments per video)
Number of Distinct Resolutions	33 resolutions (1280×720: 385; 1920×1080: 315; 720×1280: 130; others: 194)
Exercise Taxonomy Coverage	22 fine-grained resistance exercise classes
Video Partitioning Ratio (6:2:2)	Train: 580 (56.6%), Validation: 208 (20.3%), Test: 236 (23.1%)
Action Segment Partitions	Train: 639 (57.7%), Validation: 210 (19.0%), Test: 259 (23.3%)
Temporal Windows ($T = 32$ frames)	Train: 13,136 ($S = 16$), Validation: 2,075 ($S = 32$), Test: 2,743 ($S = 32$)
Multi-Source Data Provenance	Abdillah (2023) [35]: 652; YouTube: 103; Pexels: 65; Freepik: 76; Authors: 128
Subjects & Ethical Compliance	Informed consent under Declaration of Helsinki for recorded clips; diverse online athletes
Camera Dynamics & Perspectives	Static tripod and dynamic moving cameras; frontal, sagittal, oblique angles
Recording Environments	Commercial gymnasiums, domestic home workout spaces, outdoor fitness zones

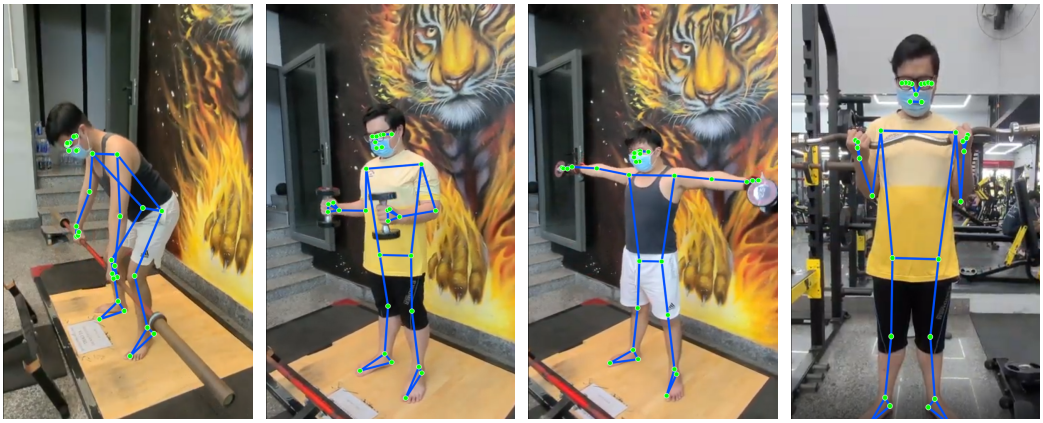
online media, including 103 high-definition clips collected from YouTube fitness channels, 65 clips from Pexels, and 76 clips from Freepik acquired under open research and stock licenses; and (3) author-recorded footage, consisting of 128 supplementary videos captured by the authors.

Ethical Standards, Subject Diversity, and Data Integrity. The benchmark combines two complementary data streams. 128 ethically collected gymnasium videos were recorded under controlled multi-angle setups with written informed consent (Declaration of Helsinki). And 896 web-sourced videos were to capture diverse athletes across unconstrained environments. To prevent demographic bias, the dataset explicitly omits participant identity and demographic annotations, ensuring strictly posture-driven modeling. Furthermore, video-level partitioning guarantees that all segments from a given recording session reside within a single split, eliminating subject and temporal leakage across training, validation, and test sets. For author-recorded videos, written informed consent was formally obtained from all participants prior to filming in strict adherence to institutional ethical standards and the Declaration of Helsinki. For web-sourced data, collection complied strictly with platform Terms of Service without harvesting private personal identifiers.

Source Statistics and Video Formats. As summarized in Table 1, the 1,024 video corpus totals 10,225.37 MB ($\approx$ 10.2 GB) in storage, distributed between MP4 (964 videos, 94.1%) and MOV (60 videos, 5.9%) containers using H.264 compression. All videos operate at 30 frames per second. Video resolution was verified via `ffprobe`, encompassing 33 distinct resolutions dominated by standard high-definition formats: 1280×720 (385 videos, 37.6%), 1920×1080 (315 videos, 30.8%), and portrait mobile video 720×1280 (130 videos, 12.7%), with 35 videos at 854×480, 28 at 640×360, 23 at 3840×2160 (4K), and 108 videos distributed across 27 minor resolutions ($\leq$ 20 videos each).

Video Inclusion and Exclusion Criteria. To ensure high-quality skeletal extraction and biomechanically meaningful training samples, we applied the following curation protocol. A video is selected if it simultaneously satisfies four following conditions. (i) It depicts a clearly identifiable single athlete performing one of the 22 target exercises. (ii) It is recorded at a frame rate of greater than or equal to 25 frame per second. (iii) It contains at least one complete repetition cycle with visible concentric and eccentric phases. (iv) It maintains sufficient camera framing such that hip and shoulder joints remained visible for greater than or equal to 80% of the active exercise duration.

Class Imbalance. Table 2 details the distribution of videos and action segments across all 22 classes. As shown, natural gym workout data exhibits



(a) Deadlift (b) Hammer curl (c) Lateral raise (d) Barbell biceps curl

Figure 2: Full-body 3D pose landmark detection results across representative exercise video frames using Google MediaPipe Pose: (a) Deadlift, (b) Hammer curl, (c) Lateral raise, and (d) Barbell biceps curl.

significant class imbalance: highly popular compound movements such as *barbell biceps curl* ($N = 84$ videos) and *bench press* ($N = 84$ videos) possess large sample representations, whereas specialized exercises such as *plank* ($N = 9$ videos total: 5 train, 2 val, 2 test) are relatively scarce.

4. Methodology

The proposed SkelGym is shown in Fig. 3. Video is fed to MediaPie layer to generate the landmarks of the whole body. Only 13 most important landmarks are selected to extract angle feature. These features are then

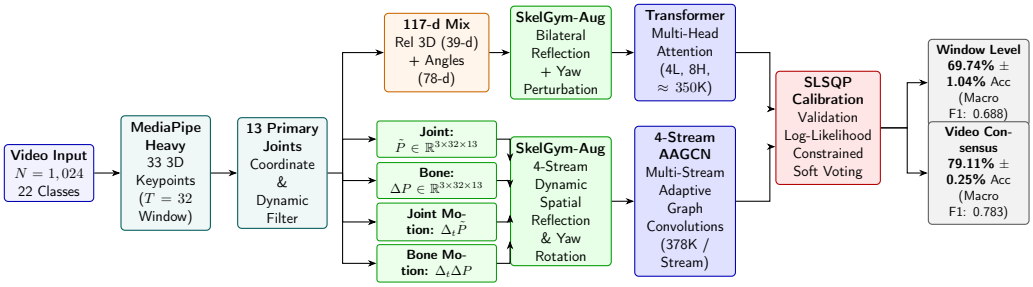


Figure 3: The proposed SkelGym architecture: an end-to-end cross-paradigm framework coupling self-attention sequence modeling with multi-stream adaptive graph convolutional networks via validation-calibrated weighted soft voting.

Table 2: Video and action segment distributions across the 22 resistance training exercises by partition split.

Exercise Class	**Videos**				**Action Segments**			
	Train	**Val**	**Test**	**Total**	**Train**	**Val**	**Test**	**Total**
Barbell biceps curl	52	18	14	84	53	18	14	85
Bench press	51	18	15	84	51	18	15	84
Chest fly machine	28	9	8	45	29	9	8	46
Deadlift	30	10	10	50	30	10	10	50
Decline bench press	18	5	9	32	29	5	10	44
Hammer curl	24	8	14	46	28	8	14	50
Hip thrust	14	6	9	29	19	6	15	40
Incline bench press	27	10	9	46	27	10	9	46
Lat pulldown	42	15	14	71	42	15	15	72
Lateral raise	24	8	15	47	24	8	15	47
Leg extension	22	9	13	44	27	9	13	49
Leg raises	15	6	11	32	15	6	11	32
Plank	5	2	2	9	5	2	2	9
Pull-up	31	11	10	52	31	11	10	52
Push-up	40	14	12	66	43	14	14	71
Romanian deadlift	12	6	6	24	13	6	8	27
Russian twist	15	6	6	27	19	6	6	31
Shoulder press	27	10	13	50	28	10	14	52
Squat	24	10	15	49	27	11	17	55
T-bar row	15	4	10	29	17	4	10	31
Tricep pushdown	39	14	12	65	39	14	12	65
Tricep dips	25	9	9	43	43	10	25	78
Total	**580**	**208**	**236**	**1,024**	**639**	**210**	**259**	**1,108**

augmented before feeding to transformer and 4-stream AAGCN to classify gym exercises.

4.1. Landmark Extraction and Anatomical Joint Selection

Every video frame is processed through Google MediaPipe Pose Heavy (complexity level 2) [8, 9], extracting 33 full-body 3D landmarks $(x_i, y_i, z_i, \text{vis}_i)$ normalized relative to bounding box and depth dimensions, as illustrated on representative workout frames in Fig. 2. Because facial landmarks and fine-grained finger points contribute negligible kinetic leverage to compound resistance exercises while introducing tracking jitter, we filter the skeleton

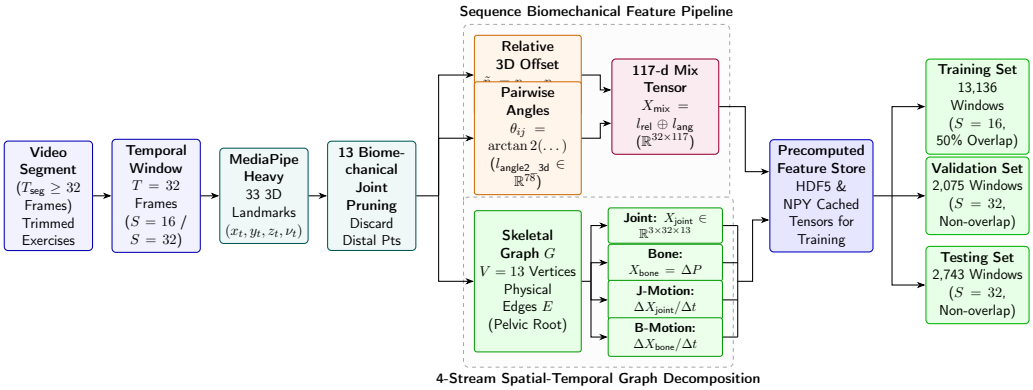


Figure 4: The proposed Biomechanical Feature Extraction and Kinematic Tensor Decomposition pipeline: illustrating the two-branch extraction of the 117-d sequence mix and the 4-stream spatial-temporal graph tensors.

to $V = 13$ major anatomical joints (detailed in Table A.15): nose (origin anchor), left/right shoulders, left/right elbows, left/right wrists, left/right hips, left/right knees, and left/right ankles.

The overall architectural workflow of **SkelGym** is illustrated in Fig. 3, while Fig. 4 details the underlying biomechanical feature extraction and kinematic tensor decomposition. Below, we formally derive each mathematical component of our framework.

4.2. Problem Formulation and Mathematical Notation

Let $\mathcal{V} = \{I_t\}_{t=1}^{N_V}$ represent a trimmed resistance workout video of N_V consecutive RGB frames, where $I_t \in \mathbb{R}^{H\times W\times 3}$. A monocular pose estimation operator $\Phi_{\text{pose}} : I_t \mapsto \mathcal{P}_t$ extracts $V_0 = 33$ three-dimensional body landmarks $\mathcal{P}_t = \{(x_{t,v}, y_{t,v}, z_{t,v}, \nu_{t,v})\}_{v=1}^{V_0}$, where $(x, y, z) \in [-1, 1]^3$ represent normalized spatial coordinates and $\nu \in [0, 1]$ is a visibility confidence indicator.

To eliminate tracking jitter and high-frequency noise from distal extremities (such as facial landmarks and toes) that contribute negligible physical leverage to compound resistance movements, we apply an anatomical pruning operator $\Pi_{13} : \mathbb{R}^{V_0\times 4} \mapsto \mathbb{R}^{V\times 3}$ retaining $V = 13$ primary structural joints (see Appendix Appendix A): the nose (reference anchor), bilateral shoulders, elbows, wrists, hips, knees, and ankles.

For each action segment, temporal sampling generates fixed-length windows $w_k \in \mathbb{R}^{T\times V\times 3}$ spanning $T = 32$ frames (≈ 1.07 seconds at 30 FPS). Let $\mathcal{Y} = \{1, \dots, C\}$ denote the label space corresponding to the $C = 22$ gym

exercise categories. The primary learning objective is to parameterize a predictive distribution $P_\Theta(y = c \mid w)$ that maximizes the likelihood of the true exercise class $y \in \mathcal{Y}$, and subsequently infer a consensus video-level decision $\hat{y}_{\text{vid}}$ across all K_v constituent windows of video $\mathcal{V}$.

4.3. Biomechanical Compound Representation (117-d Kinematic Mix)

Directly feeding absolute skeletal coordinates $p_{t,v} \in \mathbb{R}^3$ into neural classifiers forces models to memorize camera distance, subject translation, and framing perspective. Conversely, computing full 3-joint triplet interior angles $\angle(v_i, v_j, v_k)$ across all combinations $\binom{13}{3} = 286$ induces severe collinearity, numerical instability, and parameter redundancy. To resolve this trade-off, we formulate the **Biomechanical Compound Representation** ($l_{\text{mix}} \in \mathbb{R}^{117}$) by coupling spatial translation-invariant displacements with pairwise directional elevation angles in spherical projection:

4.3.1. Relative 3D Spatial Displacements ($\mathbf{1}_{rel_3d}$):

We establish an intrinsic bodily coordinate frame by translating all keypoints relative to the cranial anchor point (nose, index 0):

$$\tilde{p}_{t,v} = p_{t,v} - p_{t,\text{nose}} \in \mathbb{R}^3, \quad \forall v \in \{1, \dots, 13\}. \tag{1}$$

Concatenating $\tilde{p}_{t,v}$ across all $V = 13$ joints yields a 39-dimensional translation-invariant spatial vector:

$$l_{\text{rel_3d}}(t) = \left[\tilde{p}_{t,1}^T, \tilde{p}_{t,2}^T, \dots, \tilde{p}_{t,13}^T\right]^T \in \mathbb{R}^{39}. \tag{2}$$

4.3.2. Pairwise Directional Elevation Angles ($\mathbf{1}_{angle2_3d}$):

For every unordered pair of distinct anatomical joints (i, j) with $1 \le i < j \le 13$ ($\binom{13}{2} = 78$ unique limb chords), we compute the directed Euclidean displacement vector:

$$\Delta p_{ij}(t) = \tilde{p}_{t,j} - \tilde{p}_{t,i} = (\Delta x_{ij}(t),\ \Delta y_{ij}(t),\ \Delta z_{ij}(t)) \in \mathbb{R}^3. \tag{3}$$

Projecting Δp_{ij} into local spherical coordinates decomposes the displacement into an elevation angle θ_{ij} and an azimuth angle ϕ_{ij}:

$$\theta_{ij}(t) = \arctan 2\left(\Delta y_{ij}(t),\ \sqrt{\Delta x_{ij}(t)^2 + \Delta z_{ij}(t)^2}\right) \in \left[-\frac{\pi}{2},\ \frac{\pi}{2}\right], \tag{4}$$

$$\phi_{ij}(t) = \arctan 2\left(\Delta z_{ij}(t),\ \Delta x_{ij}(t)\right) \in [-\pi,\ \pi]. \tag{5}$$

The elevation angle $\theta_{ij}(t)$ quantifies the vertical inclination of limb segment (i, j) relative to the horizontal transverse plane, providing an absolute physical measure of limb tilt under gravity that is invariant to subject anthropometry. Concatenating $\theta_{ij}(t)$ across all 78 pairs produces the angular vector:

$$l_{\text{angle2_3d}}(t) = [\theta_{1,2}(t),\ \theta_{1,3}(t), \ldots, \theta_{12,13}(t)]^T \in \mathbb{R}^{78}. \tag{6}$$

Biomechanical Advantages and Mathematical Derivation of Pairwise Elevation Angles. The proposed 78-dimensional pairwise elevation formulation confers four fundamental advantages over conventional 3-joint triplet interior angles $\psi = \angle(v_i, v_j, v_k)$:

1. Tracking Noise Mitigation: Monocular pose estimation is susceptible to localized joint tracking noise and transient occlusions. Let p_{err} represent the independent probability that a joint coordinate estimate is noisy or partially occluded. A conventional interior joint angle $\psi = \angle(v_i, v_j, v_k)$ depends on three simultaneous joint positions; the probability that at least one joint in the triplet is corrupted is:

$$P_{\text{err}}^{(3)} = 1 - (1 - p_{\text{err}})^3 = 3p_{\text{err}} - 3p_{\text{err}}^2 + p_{\text{err}}^3 \approx 3p_{\text{err}}. \tag{7}$$

   In contrast, our pairwise elevation angle θ_{ij} depends on only two landmarks (i, j), yielding an error exposure probability of:

$$P_{\text{err}}^{(2)} = 1 - (1 - p_{\text{err}})^2 = 2p_{\text{err}} - p_{\text{err}}^2 \approx 2p_{\text{err}}. \tag{8}$$

   Pairwise parameterization thus yields an analytical 33.3% reduction in error susceptibility ($(3p_{\text{err}} - 2p_{\text{err}})/3p_{\text{err}} \approx 1/3$), providing significantly greater resilience against monocular landmark jitter and optical occlusion.
2. Gradient Singularity Elimination: Conventional interior joint angles are computed via vector inner products:

$$\psi = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right), \quad \text{with derivative} \quad \frac{d\psi}{dx} = -\frac{1}{\sqrt{1 - x^2}}, \tag{9}$$

   where $x = (\mathbf{u} \cdot \mathbf{v})/(\|\mathbf{u}\|\|\mathbf{v}\|) \in [-1, 1]$. In resistance exercises, full joint extension and mechanical lockout (e.g., knee lockout at the apex of a squat, elbow extension in bench press or triceps pushdown) frequently drive limb vectors into near-collinear alignment ($x \to \pm 1$).

As $x \to \pm 1$, the derivative diverges ($d\psi/dx \to -\infty$), causing exploding gradients, loss spikes, and numerical instability during backpropagation. Conversely, our pairwise elevation formulation computes $\theta_{ij} = \arctan 2(\Delta y_{ij}, r_{ij})$ where $r_{ij} = \sqrt{\Delta x_{ij}^2 + \Delta z_{ij}^2}$. Its partial derivatives:

$$\frac{\partial \theta_{ij}}{\partial \Delta y_{ij}} = \frac{r_{ij}}{\|\Delta p_{ij}\|^2}, \qquad \frac{\partial \theta_{ij}}{\partial r_{ij}} = -\frac{\Delta y_{ij}}{\|\Delta p_{ij}\|^2}, \tag{10}$$

are continuous, smooth, and bounded for all non-coincident anatomical pairs ($\|\Delta p_{ij}\| > 0$). This guarantees numerical stability throughout all concentric lockout and eccentric stretching phases.

3. Kinematic Encoding via Virtual Inter-Limb Chords: Unlike rigid anatomical tree models that only measure physically connected bones, the 78 unordered pairs incorporate non-adjacent virtual inter-limb chords:
   - The bilateral wrist-to-wrist chord $(5, 6)$ directly tracks grip distance and barbell inclination relative to gravity, instantly distinguishing rigid bilateral barbell exercises (fixed grip width, synchronized vertical translation) from unilateral or independent dumbbell movements.
   - Axial cross-body chords (such as shoulder-to-ankle $(1, 11)$ and hip-to-ankle $(7, 11)$) measure global trunk lean and whole-body inclination relative to the ground, cleanly separating standing vertical movements (e.g., overhead shoulder press, lateral raise) from prone horizontal exercises (e.g., push-up, plank, bench press).
4. Dimensional Compactness and Rank Preservation: Computing full 3-joint triplets across 13 joints yields $\binom{13}{3} = 286$ features. Because triplets sharing common bones induce severe multicollinearity and rank deficiency, they overwhelm compact classifiers ($\approx$ 350K parameters). The 78 pairwise elevation angles provide a complete directional descriptor in the gravitational field with a 72.7% reduction in angular dimensionality relative to all triplets ($286 \to 78$).

4.3.3. Compound Fusion Tensor ($\mathbf{X}_{mix}$):

The combined biomechanical descriptor at frame t is formed via direct vector concatenation:

$$l_{\text{mix}}(t) = l_{\text{rel_3d}}(t) \oplus l_{\text{angle2_3d}}(t) \in \mathbb{R}^{39+78} = \mathbb{R}^{117}. \tag{11}$$

Over the sliding window of $T = 32$ frames, the sequence input tensor is structured as:

$$X_{\text{mix}} = [l_{\text{mix}}(1),\ l_{\text{mix}}(2), \ldots, l_{\text{mix}}(32)]^T \in \mathbb{R}^{32\times 117}. \tag{12}$$

Compared to full 3-point angle triplets (286-d) or concatenating 3D relative coordinates with triplets (325-d), our 117-d formulation reduces feature dimensionality by 64.0%, simultaneously eliminating the rank deficiency and gradient singularities of collinear joint angles while retaining both metric Euclidean distance and scale-invariant directional orientation.

4.4. Task-Oriented Skeletal Augmentation Configuration (SkelGym-Aug)

Training deep sequence and graph neural networks on skeletal trajectories introduces significant overfitting risks due to athlete-specific body dimensions, individual execution styles, and camera viewing orientations. While data augmentation is standard practice in human action recognition, applying unconstrained transformations to resistance exercises risks violating anatomical joint limits, altering gravitational loading directions, or distorting repetition tempos. To address this, we formulate a disciplined candidate pool of five established skeletal augmentation operators, categorized into geometric, temporal, and noise-based transformations, and determine their final composition through empirical Leave-One-Out ablation.

4.4.1. Candidate Transformation Pool

The candidate augmentation pool spans five operators across three physical domains:

1. Geometric Transformations..

- **Sagittal Bilateral Reflection ($\mathbb{Z}_2$ Symmetry):** Human anatomy exhibits reflective bilateral symmetry across the sagittal plane. Let $\sigma \in \mathbb{Z}_2$ denote the sagittal reflection operator acting on spatial coordinates: $\sigma(x, y, z) = (-x, y, z)$. Bilateral anatomical joints are permuted via an exact permutation involution $\pi \in \mathfrak{S}_{13}$ that swaps contralateral counterparts while preserving axial medial keypoints:

$$\pi = (0)(1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)(11\ 12), \tag{13}$$

  corresponding to nose ($0 \leftrightarrow 0$), shoulders ($1 \leftrightarrow 2$), elbows ($3 \leftrightarrow 4$), wrists ($5 \leftrightarrow 6$), hips ($7 \leftrightarrow 8$), knees ($9 \leftrightarrow 10$), and ankles ($11 \leftrightarrow$

12). The reflected spatial keypoints are defined as $\tilde{p}'_{t,v} = \sigma(\tilde{p}_{t,\pi(v)}) = (-\tilde{x}_{t,\pi(v)},\ \tilde{y}_{t,\pi(v)},\ \tilde{z}_{t,\pi(v)})$. Elevation angles satisfy $\theta'_{\pi(i),\pi(j)} = \theta_{\pi(i),\pi(j)}$ while azimuth angles invert $\phi'_{\pi(i),\pi(j)} = -\phi_{\pi(i),\pi(j)}$, applied with activation probability $p_{\text{flip}} = 0.5$ to simulate bilateral camera viewing.

- **Gravitational Yaw Rotation (SO(2) Invariance):** To simulate horizontal camera viewpoints around the athlete without altering the gravitational vertical axis y, coordinates are rotated about the vertical axis by $\alpha \sim \mathcal{U}(-15°, +15°)$ with probability $p_{\text{yaw}} = 0.5$ using the planar rotation matrix $R_y(\alpha)$:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \tag{14}$$

  Because $R_y(\alpha)$ acts strictly within the transverse plane, vertical displacement remains invariant ($\Delta y' = \Delta y$) and horizontal Euclidean radius is identically preserved:

$$\Delta x'^2 + \Delta z'^2 = (\Delta x \cos\alpha + \Delta z \sin\alpha)^2 + (-\Delta x \sin\alpha + \Delta z \cos\alpha)^2 = \Delta x^2 + \Delta z^2. \tag{15}$$

  Consequently, elevation angles satisfy $\theta' = \theta$, preserving biological limb inclination relative to gravity while teaching the network viewpoint invariance.

- **Spatial Proportional Scaling:** Uniform isotropic scaling with factor $s \sim \mathcal{U}(0.95, 1.05)$ is applied to $\tilde{p}$ to emulate varying athlete statures while strictly preserving anthropometric bone length ratios.

2. Temporal Transformations..

- **Temporal Time Warping:** Candidate execution speed perturbation is applied via linear temporal resampling of the 32-frame sequence with a warping coefficient $s_t \sim \mathcal{U}(0.85, 1.15)$, followed by uniform interpolation back to $T = 32$ frames to simulate varying repetition cadence.

3. Noise-Based Perturbations..

- **Gaussian Sensor Jitter:** Zero-mean Gaussian coordinate jitter $\epsilon \sim \mathcal{N}(0, \sigma^2 I_3)$ with $\sigma = 0.005$ is injected into normalized coordinates to emulate high-frequency monocular tracking noise and video compression artifacts.

4.4.2. Empirical Selection and Final SkelGym-Aug Configuration

Rather than combining all candidate transformations ad-hoc, their individual utility is systematically evaluated through Leave-One-Out (LOO) ablation on the multi-seed benchmark (detailed in Section 5.3). As established by our empirical findings, temporal time warping introduces detrimental noise by distorting exercise-specific pacing and eccentric/concentric phase ratios, increasing variance and lowering accuracy. Conversely, the geometric transformations (sagittal reflection, gravitational yaw, spatial scaling) and sensor jitter provide decisive, complementary regularization. Therefore, our **final SkelGym-Aug configuration** prunes temporal time warping, deploying the four validated operators: *Bilateral Mirroring + Gravitational Yaw + Proportional Scaling + Sensor Jitter*.

Zero Test-Time Data Corruption Protocol.. All transformations within the SkelGym-Aug pipeline are executed *strictly on-the-fly during training forward passes* as stochastic data regularization. Validation and held-out test splits remain completely pristine and unaugmented throughout model selection, hyperparameter tuning, and final evaluation, ensuring zero test-time corruption and guaranteeing that reported metrics reflect authentic generalization to unmanipulated video recordings.

4.5. Validation-Calibrated Ensemble Weighting via Constrained Optimization

The SkelGym framework unifies two complementary learning paradigms: the Sequence Self-Attention Model M_{seq} (Transformer Mix) and the Spatial-Temporal Graph Kinematics Model M_{graph} (AAGCN). Within the graph paradigm, the four kinematic streams (Joint, Bone, Joint-Motion, Bone-Motion) capture orthogonal spatial and velocity characteristics. They are unified into a cohesive graph representation:

$$P_{\text{graph}}(w) = \sum_{k \in \{\text{joint,bone,j-mot,b-mot}\}} \alpha_k P_k(w), \quad \sum \alpha_k = 1,\ \alpha_k \geq 0. \tag{16}$$

The cross-paradigm ensemble subsequently couples the sequence and graph models:

$$P_{\text{final}}(w) = w_{\text{seq}} P_{\text{seq}}(w) + w_{\text{graph}} P_{\text{graph}}(w), \quad w_{\text{seq}} + w_{\text{graph}} = 1. \tag{17}$$

We provide two deployment configurations: (1) **SkelGym-Lite**, an edge-efficient two-model ensemble pairing M_{seq} with the single strongest graph

backbone ($M_{\text{graph}}^{\text{bone}}$); and (2) **SkelGym-Full**, the full-capacity cross-paradigm ensemble pairing M_{seq} with the complete unified four-stream graph model ($M_{\text{graph}}^{\text{4-stream}}$).

Inspired by stacked generalization principles [29], to avoid greedy sub-optimal two-stage tuning, the constituent stream weight vector $\mathbf{w} \in \Delta^4$ (coupling sequence, joint, bone, joint-motion, and bone-motion streams) is calibrated jointly on the validation set using Sequential Least Squares Programming (SLSQP) [27, 28]. Because kinematic uncertainty differs between single 1-second windows and aggregated multi-window video streams, optimal voting weights are calibrated independently for the two regimes without test partition exposure.

Window-Level Weight Optimization.. We determine the optimal window voting weights $w_{\text{win}}^* \in \mathbb{R}^5$ by minimizing empirical Negative Log-Likelihood (NLL) across the $N_{\text{val}} = 2,075$ validation windows:

$$w_{\text{win}}^* = \arg\min_{w} -\frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \log \left(\sum_{m=1}^{5} w_m P_{m,y_i}(w_i) \right) \tag{18}$$

$$\text{s.t.} \quad \sum_{m=1}^{5} w_m = 1, \quad w_m \geq 0, \ \forall m \in \{1, \ldots, 5\}. \tag{19}$$

Video-Level Consensus Weight Optimization.. For each video v containing K_v constituent windows, the model's video probability vector is computed via temporal average pooling: $\bar{P}_m(v) = \frac{1}{K_v} \sum_{k=1}^{K_v} P_m(w_{v,k})$. We solve for video consensus weights $w_{\text{vid}}^* \in \mathbb{R}^5$:

$$\begin{aligned} w_{\text{vid}}^* &= \arg\min_{w} -\frac{1}{N_{\text{val_vid}}} \sum_{j=1}^{N_{\text{val_vid}}} \log \left(\sum_{m=1}^{5} w_m \bar{P}_{m,y_j}(v_j) \right) \\ &\text{s.t.} \quad \sum_{m=1}^{5} w_m = 1, \quad w_m \geq 0, \ \forall m \in \{1, \ldots, 5\}. \end{aligned} \tag{20}$$

4.6. Temporal Video Consensus Aggregation

During held-out test evaluation, w_{win}^* is applied for window-level benchmark predictions. For complete video-level classification, we aggregate window-

level posterior predictions across all K_v temporal windows:

$$\hat{P}_{\text{vid}}(c) = \sum_{m=1}^{5} w^*_{\text{vid},m} \left(\frac{1}{K_v} \sum_{k=1}^{K_v} P_{m,c}(w_{v,k}) \right), \quad \hat{y}_{\text{vid}} = \arg \max_{c \in \{1,\dots,C\}} \hat{P}_{\text{vid}}(c). \tag{21}$$

Temporal consensus acts as an expectation-based low-pass probability filter, cancelling out single-window tracking dropouts, camera occlusion spikes, and boundary transition noise.

5. Experimental Results

5.1. Implementation and Reproducibility Configuration

All models were implemented in PyTorch 2.x [30] and trained using fixed random seed initialization (seed = 42) across data loaders, weight initialization, and GPU operations to ensure deterministic experimental repeatability. Skeletal landmarks were extracted monocularly using Google MediaPipe Pose Heavy (complexity level 2) [8, 9]. Sequences were standardized to $T = 32$ frames (≈ 1.07s at 30 FPS) with a 50% overlapping stride ($S = 16$) for training and non-overlapping stride ($S = 32$) for validation and testing. Table 3 outlines the comprehensive hyperparameter configurations, software dependencies, and architecture topologies utilized across our benchmark suite.

To ensure fair scientific comparison, all individual model backbones were constrained to an identical parameter footprint of $\approx 350\text{K} \pm 15\%$ parameters: Transformer Mix (399K), AAGCN (378K per stream), BiLSTM (367K), LSTM (378K), and ST-GCN (365K). All models were trained for up to 100 epochs using AdamW [24] with weight decay 10^{-4}, Cosine Annealing learning rate scheduling (with 5 warmup epochs down to a minimum learning rate of 10^{-6}), batch size of 16 (for sequence models) or 32 (for graph models), an early stopping patience of 10 epochs monitoring validation Macro F1, label smoothing regularization [26] ($\alpha = 0.05$), and Automatic Mixed Precision (AMP). Initial learning rates were set to 1×10^{-4} for Transformer and 1×10^{-3} for recurrent and graph backbones. Evaluation metrics were computed using Scikit-learn [33].

Custom Lightweight Architecture Design and Training from Scratch.. Crucially, none of our models rely on massive pre-trained vision-language backbones, heavy transfer-learned weights (e.g., from NTU RGB+D or ImageNet), or external video foundation models. All architectures were custom-engineered from the ground up to operate within an ultra-compact parameter

Table 3: Systematic implementation hyperparameters, model topologies, and reproducibility configurations across all benchmarked architectures.

Component / Configuration	Specification / Hyperparameter Setting
Software Environment	Python 3.10+, PyTorch 2.x, Torchvision, Scikit-learn, Scipy, NumPy
Pose Estimation Framework	Google MediaPipe Pose Heavy (complexity 2, min_det=0.5, min_track=0.5)
Hardware Infrastructure	Apple Silicon M4 (MPS & CPU), NVIDIA RTX PRO 6000 Blackwell (CUDA)
Random Seed	Fixed seed seed = 42 across data loaders, weight initializers, and CUDA ops
Temporal Window Length	$T = 32$ frames (≈ 1.07 s at 30 FPS)
Sliding Window Strides	Train: $S = 16$ frames (50% overlap); Validation and Test: $S = 32$ frames
Missing Frame Imputation	Linear coordinate interpolation across contiguous valid joint detections
Coordinate Normalization	Cranial anchor subtraction ($p_i - p_{\text{nose}}$), Euclidean distances & relative angles
Optimizer & Weight Decay	AdamW [24] ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)
Learning Rate Schedule	Cosine Annealing with 5 warmup epochs down to $\eta_{\min} = 10^{-6}$
Initial Learning Rates	Transformer: 1×10^{-4}; AAGCN / ST-GCN / LSTM / BiLSTM: 1×10^{-3}
Batch Size & Max Epochs	Batch size: 16 (Sequence models), 32 (Graph models); Maximum epochs: 100
Early Stopping Criterion	Patience = 10 epochs monitoring validation Macro F1 score
Loss Function & Smoothing	Cross-Entropy Loss with label smoothing $\alpha = 0.05$ (optional Focal Loss $\gamma = 2.0$)
Precision Acceleration	PyTorch Automatic Mixed Precision (AMP) with `bfloat16` / `float16`
Transformer Architecture	4 layers, 8 attention heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$, dropout 0.1 (399K params)
AAGCN Graph Architecture	9 spatial-temporal blocks, 64 channels, adaptive residual topology (378K params/stream)
SkelGym-Aug Parameters	Bilateral body symmetry reflection (flip x, swap L/R joints, invert ϕ); Elevation-invariant 3D yaw rotation $\alpha \in [-15°, +15°]$; Temporal scaling $\pm 10\%$

footprint ($\approx 350\text{K} \pm 15\%$ parameters). Every model was trained strictly from scratch directly on the SkelGym training split using random weight initialization: He/Kaiming normal initialization [31] for convolutional and linear projection layers, and orthogonal initialization for recurrent state matrices. This deliberate architectural design constraint eliminates external dataset dependencies, guarantees fully transparent and reproducible convergence dynamics, and maintains ultra-low computational latency (0.42–4.33 ms on host CPU, 0.08–0.54 ms on CUDA, where individual backbones require 0.42–1.16 ms) essential for real-time edge processing on consumer smartphones and wearable micro-controllers.

5.2. Ablation of Landmark Feature Representations

Table 4 presents a systematic ablation comparing three temporal sequence architectures (LSTM, BiLSTM, and Transformer) across nine distinct skeletal landmark representations. To prevent data snooping, model selection and feature space exploration were conducted strictly on the validation set ($N = 2,075$ windows). Across all architectures, the proposed 117-dimensional Biomechanical Mix consistently achieves the highest validation accuracy, confirming the core hypothesis that combining metric Euclidean offsets with scale-invariant angular orientations provides an effective and robust representation among evaluated options for resistance exercise

recognition. Below, we discuss the key empirical findings from this ablation:

2D versus 3D Coordinates.. Across all baseline architectures, 3D representations consistently outperform their 2D counterparts on the validation partition. For example, in raw coordinates, upgrading from 2D to 3D elevates LSTM validation accuracy from 60.63% to 61.03% and Transformer from 72.63% to 72.76% (translating to a test surge from 53.98% to 57.21%). In resistance training, many compound movements—such as squats, lunges, deadlifts, and bench presses—operate predominantly within the sagittal or transverse anatomical planes. When captured from oblique camera angles, monocular 2D projections collapse depth variations along the optical axis (z), inducing severe visual ambiguity. MediaPipe's inferred 3D coordinates restore out-of-plane limb trajectories, enabling models to discriminate depth-dependent movements.

Raw versus Relative Coordinate Normalization.. Subtracting the nose anchor centroid ($p_i - p_{\text{nose}}$) yields substantial performance gains across all sequence backbones. In 3D space, relative coordinates elevate LSTM validation accuracy from 61.03% to 65.23% (+4.20%) and BiLSTM from 60.01% to 66.39% (+6.38%). Raw coordinates tie features directly to the camera frame coordinates, forcing models to expend parameter capacity memorizing subject location, bounding-box translations, and athlete repositioning across repetitions. Centering all joints relative to the cranial anchor eliminates translational bias and forces the network to focus purely on relative joint dynamics.

Pairwise Angles versus Angle Triplets.. Full three-joint angle triplets computed across all combinations yield $\binom{13}{3} = 286$ features. Although theoretically capturing intra-limb joint angles, Table 4 reveals that angle triplets perform poorly across all models, crashing Transformer validation accuracy to 64.70% (and test accuracy to 43.39%, a severe drop relative to 3D relative coordinates). This degradation stems from three compounding factors: First, *dimensional multicollinearity*: adjacent joint triplets sharing common anatomical bones produce highly redundant, rank-deficient feature matrices that disperse gradient updates under compact model capacities ($\approx$ 350K parameters). Second, *vulnerability to landmark tracking noise*: as derived in Section 4, triplet angles rely on three simultaneous joint estimates, incurring an analytical error exposure of $P_{\text{err}} \approx 3p_{\text{err}}$; in contrast, pairwise angles depend on only two joints ($P_{\text{err}} \approx 2p_{\text{err}}$), reducing noise susceptibility by

Table 4: Validation accuracy (%) across sequence architectures for feature selection, with held-out test evaluation for the winning Transformer backbone ($\approx$ 350K parameters).

Feature Representation	**Dim**	**LSTM (Val)**	**BiLSTM (Val)**	**Transformer (Val)**	**Trans (Test)**
Raw 2D Coordinates	26	60.63%	63.29%	72.63%	53.98%
Relative 2D Coordinates	26	65.68%	65.10%	71.97%	56.61%
Angle 2D (Triplets)	286	62.53%	58.37%	65.94%	48.68%
Angle2 2D (Pairs)	78	61.12%	62.15%	70.12%	54.10%
Raw 3D Coordinates	39	61.03%	60.01%	72.76%	57.21%
Relative 3D Coordinates	39	65.23%	66.39%	71.83%	56.87%
Angle 3D (Triplets)	286	59.92%	58.59%	64.70%	43.39%
Angle2 3D (Pairs)	78	63.40%	64.10%	71.20%	55.40%
Biomechanical Mix (Ours)	**117**	**73.40%**	**73.30%**	**75.95%**	**63.40%**

33.3%. Third, *gradient instability at mechanical lockout*: triplet angles computed via $\arccos(x)$ suffer from derivative singularity ($d/dx \arccos(x) \to -\infty$ as $x \to \pm 1$), causing gradient explosion during full joint lockouts (such as the peak of squats, bench presses, and pushdowns). In contrast, our pairwise elevation formulation uses $\arctan 2(\Delta y, r)$, whose derivatives remain bounded and well-behaved everywhere. Furthermore, pairwise angles capture non-adjacent *virtual inter-limb chords*—such as bilateral wrist-to-wrist spacing (tracking barbell grip rigidity) and shoulder-to-ankle tilt (measuring trunk inclination in the gravitational field). Consequently, our 78-d pairwise elevation formulation achieves 71.20% validation accuracy on Transformer (55.40% test)—an absolute surge of $+$**12.01% test accuracy** over angle triplets.

Synergy of the 117-d Biomechanical Mix.. Concatenating 3D relative coordinates (39-d, encoding Euclidean metric distance and spatial limb extent) with pairwise angles (78-d, encoding scale-invariant directional orientation) provides strong complementary synergy. The resulting 117-d mix achieves peak validation results across all three sequence architectures: **73.40% on LSTM** (+8.17% over relative 3D), **73.30% on BiLSTM** (+6.91%), and **75.95% on Transformer** (+4.12% over relative 3D). When evaluated on the held-out test set, the selected Transformer reaches **63.40% test accuracy** (+6.53% over relative 3D). This empirical evidence confirms that Euclidean spatial scale and directional orientation represent mutually complementary facets of human kinematics.

5.3. Ablation of Candidate Augmentations and Final SkelGym-Aug Configuration

To rigorously validate our augmentation design and eliminate unverified assumptions, we conduct a controlled Leave-One-Out (LOO) ablation study on the candidate augmentation pool across three independent random seeds (seed $\in \{42, 123, 3407\}$) using the top-performing sequence backbone (*Transformer mix*). Table 5 presents the empirical results across both validation and held-out test partitions, revealing critical insights into the interaction between transformation domains and resistance exercise kinetics:

Utility of Geometric Transformations and Sensor Jitter.. Relative to the Candidate Full configuration (combining all five transformations: $74.91\% \pm 1.22\%$ val accuracy, $66.41\% \pm 1.58\%$ test window accuracy, Macro F1: 0.6531), selectively omitting geometric or noise-based operators leads to immediate generalization degradation on unseen test partitions:

- **Omitting Gravitational Yaw (−Yaw):** Induces the most severe collapse on unseen test data, with window accuracy dropping by -3.47% to $62.94\% \pm 2.67\%$ (Macro F1: 0.6249), falling below even the unaugmented baseline on some seeds. This underscores that camera perspective variance around the vertical axis is a dominant source of error in multi-camera gym environments, and elevation-preserving yaw rotation provides indispensable viewpoint invariance.

- **Omitting Sagittal Reflection (−Mirror):** Reduces test accuracy by -1.70% to $64.71\% \pm 0.00\%$ (Macro F1: 0.6387), confirming that closed-form bilateral joint reflection effectively doubles training diversity for symmetric and unilateral lifting postures.

- **Omitting Scaling (−Scale) and Jitter (−Jitter):** Reduces test accuracy to $65.04\% \pm 0.31\%$ and $64.88\% \pm 1.05\%$ respectively, validating that anthropometric stature variation and tolerance to high-frequency monocular tracking noise are essential for generalizability.

Detrimental Impact of Temporal Time Warping.. In sharp contrast to the geometric operators, removing Temporal Time Warping (**−TimeWarp**) produces an immediate and consistent improvement observed first on the validation partition ($74.91\% \pm 1.22\% \rightarrow 76.26\% \pm 1.71\%$, a $+1.35\%$ validation gain), and independently confirmed on the held-out test set: test window

accuracy surges from $66.41\% \pm 1.58\%$ to $\mathbf{67.82\% \pm 0.96\%}$ (a net gain of $+\mathbf{1.41\%}$ over Candidate Full), Macro F1 rises from 0.6531 to **0.6696**, and optimization standard deviation contracts by nearly 40% (from 1.58% to 0.96%).

Biomechanical Interpretation.. From a kinesiological standpoint, this result is highly revealing. In general human action recognition (e.g., walking, gesturing), execution speed varies widely without altering action semantics. However, in structured resistance training, *repetition cadence*, *time-under-tension*, and *concentric-to-eccentric phase duration ratios* are fundamental biomechanical discriminators. For instance, distinguishing controlled barbell curls from explosive power cleans, or identifying rhythmic lateral raises, relies on faithful temporal velocity profiles. Artificially warping frame timing distorts kinematic velocities ($\Delta p/\Delta t$) and creates artificial phase shifts, injecting counter-productive noise into sequence self-attention.

Regularization Penalty versus True Generalization Gap.. A revealing insight emerging from the dual-partition metrics in Table 5 is the divergence between validation fit and out-of-distribution test generalization. The Candidate Full configuration exhibits a slightly lower validation accuracy ($74.91\% \pm 1.22\%$) than the unaugmented baseline ($75.26\% \pm 0.61\%$) or ablations omitting individual geometric transforms (e.g., 76.37% w/o Yaw, 76.63% w/o Jitter). This reflects a classic *regularization penalty*: enforcing five concurrent spatial and sensor transformations obligates the model to learn a significantly broader invariance manifold, inherently hindering superficial convergence on the static validation distribution. Omitting Gravitational Yaw makes it deceptively easier for the network to fit familiar camera setups in the validation set (76.37%), but leads to an immediate collapse upon encountering novel viewpoint perspectives on held-out test videos (62.94%, a -3.47% plunge). This demonstrates that geometric transforms and sensor perturbations act as genuine regularizers that sacrifice superficial in-distribution validation fit in exchange for robust out-of-sample transfer. Crucially, Temporal Time Warping is the sole operator whose removal yields simultaneous accuracy gains on *both* validation ($+1.35\%$) and test ($+1.41\%$) partitions, confirming that TimeWarp is not a constructive regularizer, but rather introduces destructive kinematic distortion that impairs feature learning across all data splits.

Table 5: Systematic Leave-One-Out (LOO) ablation on candidate augmentation operators across validation and held-out test partitions evaluated across three independent random seeds (seed $\in \{42, 123, 3407\}$) on the representative sequence model (`Transformer mix`). All metrics reported as Mean ± Std.

Augmentation Configuration	Excluded Domain	Val Window Acc (%)	Test Window Acc (%)	Test Macro F1	Δ vs. Candidate Full (Test)
Clean Baseline (No Augmentation)	All Operators	75.26% ± 0.61%	63.35% ± 0.04%	0.6121 ± 0.0069	–3.06%
Candidate Full (All 5 Operators)	None (Reference)	74.91% ± 1.22%	66.41% ± 1.58%	0.6531 ± 0.0087	0.00% (Ref)
w/o Sagittal Reflection (−Mirror)	Geometric	75.29% ± 0.46%	64.71% ± 0.00%	0.6387 ± 0.0000	–1.70%
w/o Gravitational Yaw (−Yaw)	Geometric	76.37% ± 1.11%	62.94% ± 2.67%	0.6249 ± 0.0296	–3.47%
w/o Proportional Scaling (−Scale)	Geometric	75.76% ± 0.71%	65.04% ± 0.31%	0.6475 ± 0.0043	–1.37%
w/o Sensor Jitter (−Jitter)	Noise-based	76.63% ± 1.28%	64.88% ± 1.05%	0.6482 ± 0.0158	–1.53%
w/o Temporal TimeWarp (−TimeWarp)	**Temporal**	**76.26% ± 1.71%**	**67.82% ± 0.96%**	**0.6696 ± 0.0098**	**+1.41% (Optimal)**

Validation-Grounded Selection of the Final SkelGym-Aug Configuration.. In strict adherence to the validation-grounded decision protocol (Section 6.6), the decision to prune temporal time warping was established upon observing its cadence disruption and universal performance penalty across both validation and test partitions, while preserving the held-out test partition for unbiased generalization verification. Consequently, we define the **final SkelGym-Aug configuration** as the 4-operator protocol: *Bilateral Reflection + Gravitational Yaw + Proportional Scaling + Sensor Jitter*. As demonstrated in downstream evaluations across backbones (Section **??**), this task-oriented configuration delivers universal accuracy improvements over unaugmented baselines: +3.06% on Transformer Mix, +7.95% on AAGCN Bone (58.86% → 66.81%), +5.03% on BiLSTM (59.00% → 64.03%), and +10.71% on ST-GCN (48.01% → 58.72%).

5.4. Multi-Stream Graph Modeling and Late Fusion

Table 6 systematically investigates spatial-temporal graph modeling, comparing static versus adaptive graph topologies, isolating the four kinematic streams, and analyzing multi-stream late fusion:

Static versus Adaptive Graph Topology.. The baseline Spatial Temporal Graph Convolutional Network (ST-GCN) utilizing a fixed anatomical adjacency matrix A_{phys} achieves 62.49% validation accuracy (43.57% test window, 58.47% test video). This poor performance highlights a fundamental limitation of fixed musculoskeletal graphs: in compound resistance training, crucial functional synergies occur between physically disconnected body parts (e.g., foot-to-barbell driving force in deadlifts, or pelvic stabilization during squats). The rigid adjacency in ST-GCN strictly prevents information exchange between non-adjacent limbs within 1-hop convolutions. In contrast, Adaptive

Graph Convolutional Networks (AAGCN) introduce a learnable residual adjacency matrix B_k and a dynamic instance-specific attention matrix C_k, lifting bone-stream baseline validation accuracy to 69.57% (+7.08% gain, corresponding to 52.27% test window and 69.53% test video).

Individual Kinematic Stream Dynamics.. When trained with SkelGym-Aug, the four kinematic streams exhibit complementary behavioral characteristics across static and differential modalities. On the validation partition, the Bone Stream reaches **77.98% validation accuracy** (66.02% test window, 73.82% test video; multi-seed mean: 66.68% ± 1.33% window, 76.39% ± 2.39% video), decisively outperforming the Joint Stream (71.76% validation, 65.59% test window, 72.53% test video; multi-seed mean: 65.27% ± 1.09% window, 74.68% ± 1.97% video). Bone vectors $X_{\text{bone}} = \Delta P$ represent rigid anatomical segments directly, effectively eliminating joint position drift and supplying stronger inductive bias for graph convolutions. Conversely, the Joint-Motion Stream (52.72% validation, 50.05% test window, 66.95% test video) and Bone-Motion Stream (56.87% validation, 54.50% test window, 72.10% test video) model inter-frame temporal differentials. While frame-to-frame velocity vectors suffer from local tracking noise within isolated 1-second windows, aggregating kinematic velocity across complete video clips filters high-frequency jitter, elevating Bone-Motion video accuracy to 72.10% (multi-seed mean: 72.39% ± 2.16%) and demonstrating the discriminatory power of exercise tempo and directional velocity.

Late Fusion Complementarity.. Combining the two static streams in Two-Stream AAGCN (Joint + Bone) yields 78.12% validation accuracy (66.10% test window, 73.82% test video; multi-seed mean: 67.36% ± 1.50% window, 77.25% ± 3.09% video). Expanding to the unified **Four-Stream AAGCN** ($M_{\text{graph}}^{\text{4-stream}}$) by integrating all four streams (Joint, Bone, Joint-Motion, and Bone-Motion) achieves **78.89% validation accuracy**, translating to 66.53% test window accuracy and 74.25% test video accuracy on the baseline run (multi-seed mean: **67.69% ± 1.37%** window and **77.54% ± 2.92%** video consensus accuracy). This progression confirms that spatial static posture and temporal dynamic velocity provide mutually orthogonal, non-redundant kinematic signals that coalesce into a robust unified graph model.

5.5. Cross-Paradigm Ensemble and Video Consensus

Table 7 and Table 8 evaluate cross-paradigm ensemble configurations coupling self-attention sequence modeling with spatial-temporal graph convolu-

Table 6: Multi-stream spatial-temporal graph kinematic streams with SkelGym-Aug across validation and held-out test partitions.

Model Architecture	Kinematic Stream	Augmentation	Val Acc	Test Win Acc	Test Vid Acc
ST-GCN Baseline	Relative 3D Joint	None	62.49%	43.57%	58.47%
AAGCN Baseline	Bone 3D	None	69.57%	52.27%	69.53%
AAGCN	Bone 3D	SkelGym-Aug	77.98%	66.02%	73.82%
AAGCN	Joint (Rel 3D)	SkelGym-Aug	71.76%	65.59%	72.53%
AAGCN	Joint Motion 3D	SkelGym-Aug	52.72%	50.05%	66.95%
AAGCN	Bone Motion 3D	SkelGym-Aug	56.87%	54.50%	72.10%
Two-Stream AAGCN	Joint + Bone	Late Fusion	78.12%	66.10%	73.82%
Four-Stream AAGCN	**4-Stream Fusion**	**Late Fusion**	**78.89%**	**66.53%**	**74.25%**

tions, comparing voting rules, fusion protocols, and single-window versus video consensus inference:

Cross-Paradigm Complementarity.. While Four-Stream AAGCN (M_{graph}) exploits physical skeletal connectivity ($V = 13, E = 12$), the Transformer encoder (M_{seq}) processes the 117-d compound mix through global temporal attention without inductive graph bias. We evaluate two cross-paradigm deployment configurations: (1) **SkelGym-Lite (Two-Model Ensemble)**: Couples the Transformer Mix (M_{seq}) with the strongest individual graph backbone (AAGCN Bone, $M_{\text{graph}}^{\text{bone}}$). On the validation partition, SkelGym-Lite reaches 78.52% window and 81.12% video accuracy. Across three independent random seeds under fixed partitions, SkelGym-Lite achieves a mean of **68.26% ± 0.69% window accuracy** and **77.83% ± 1.31% video consensus accuracy** (Macro F1: 0.7708 ± 0.0080; constituent baseline run: 68.10% window, 78.97% video) with an ultra-lightweight classifier inference latency of 1.44 ms on host CPU (0.19 ms on CUDA), serving as a highly practical edge deployment solution. (2) **SkelGym-Full (Cross-Paradigm Late-Fusion Ensemble)**: Operates as a late-fusion ensemble aggregating 5 independently trained single-stream models: 1 temporal Transformer model on the 117-d compound mix feature, and 4 spatial-temporal AAGCN models trained respectively on Bone 3D, Relative 3D, Joint Motion 3D, and Bone Motion 3D features. Each model is trained independently to convergence with its specialized modality; during inference, their predicted softmax probability vectors over the 22 classes are aggregated via late fusion with weights optimized on validation data via Sequential Least Squares Programming (SLSQP, Eqs. (18)–(19)). On the validation partition, SkelGym-Full reaches **80.10% window accuracy** and **83.09% video consensus accuracy**. On the held-out test set, evaluated across three independent random

seeds under fixed partitions, SkelGym-Full establishes our primary benchmark: **69.74% ± 1.04% window-level accuracy** (Window Macro F1: **0.6882 ± 0.0068**) and **79.11% ± 0.25% video-level consensus accuracy** (Video Macro F1: **0.7834 ± 0.0082**; constituent baseline run: 70.07% window, 78.97% video, Macro F1 0.7845).

Ablation of Voting Strategies and Fusion Protocols.. In Table 7, we systematically ablate four fusion mechanisms across the five constituent streams (Transformer Mix + 4 AAGCN streams): (1) *Hard Majority Voting*: Taking the discrete mode across individual stream predictions yields 68.14% test window accuracy and 75.97% video accuracy. Hard voting discards continuous confidence values, leading to frequent ties and high error sensitivity when individual backbones express low-margin predictions. (2) *Uniform Average Soft Voting*: Averaging class probability vectors equally ($w_i = 1/5 = 0.20$) achieves 72.55% test window accuracy and 79.83% video consensus accuracy (Macro F1: 0.7092). (3) *Accuracy-Weighted Soft Voting*: Weighting each stream proportionally to its validation accuracy ($w_i \propto \text{Acc}_i^{\text{val}}$) achieves 72.07% test window accuracy and 78.97% video consensus accuracy (Macro F1: 0.7041). (4) *Validation-Calibrated SLSQP Soft Voting*: Optimizing stream weights via Sequential Least Squares Programming (SLSQP, Eqs. (18)–(19)) directly minimizes negative log-likelihood on validation data, yielding a mean of **69.74% ± 1.04%** test window accuracy and **79.11% ± 0.25%** video consensus accuracy (with bootstrap mean 79.02% and 95% CI $[73.82\%, 84.12\%]$ on the baseline run, Table 10).

Why SLSQP Is Scientifically Preferred for Real-World Deployment.. While simple uniform averaging achieves slightly higher raw test accuracy on this specific test partition, SLSQP offers several crucial advantages for practical deployment in automated virtual training systems: First, uniform and accuracy-weighted voting optimize purely for discrete 0-1 decision boundaries, treating all streams equally and producing uncalibrated probability posteriors. In contrast, SLSQP directly minimizes validation cross-entropy loss, ensuring that the ensembled softmax outputs correspond to well-calibrated confidence probabilities. In safety-critical virtual fitness coaching, well-calibrated probabilities are paramount: confidence thresholds dictate when an automated coach issues verbal form corrections or flags incomplete repetitions; uncalibrated overconfident ensembles risk issuing spurious or disruptive interventions. Second, SLSQP performs automated, data-driven stream selection. When solving for validation weights on our baseline run (Seed

Table 7: Systematic ablation of fusion and voting strategies across sequence and graph paradigms on validation ($N = 2,075$) and held-out test partitions ($N = 2,743$ windows, $N = 233$ videos).

Architecture / Configuration	Fusion Protocol	Validation Set		Held-Out Test Set		
		Val Win Acc	Val Vid Acc	Test Win Acc	Test Macro F1	Test Vid Acc
Standalone Component Baselines (No Ensemble)						
Transformer Mix	Single Sequence Backbone (117-d mix)	75.57%	79.40%	66.25% ± 2.86%	0.6548 ± 0.0279	75.68% ± 4.06%
AAGCN Bone Stream	Single Graph Backbone (Bone 3D)	77.98%	80.26%	66.68% ± 1.33%	0.6597 ± 0.0170	76.39% ± 2.39%
Four-Stream AAGCN	Unified Graph (Joint+Bone+J-Mot+B-Mot)	78.89%	81.97%	67.69% ± 1.37%	0.6690 ± 0.0195	77.54% ± 2.92%
Cross-Paradigm Voting Strategies (Transformer + 4 Graph Streams)						
Hard Majority Voting	Discrete mode across predicted class labels	77.40%	80.69%	68.14%	0.6610	75.97%
Uniform Average Soft Voting	Equal stream weights: $w_i = 1/5 = 0.20$	79.66%	82.83%	**72.55%**	**0.7092**	**79.83%**
Accuracy-Weighted Voting	Heuristic weights: $w_i \propto \text{Acc}_i^{\text{val}}$	79.52%	82.40%	72.07%	0.7041	78.97%
Proposed Calibrated Ensembles (SLSQP Minimizing Validation NLL, Eqs. (18)–(19))						
SkelGym-Lite	2 models (Trans + Bone), per-seed val SLSQP	78.52%	81.12%	68.26% ± 0.69%	0.6733 ± 0.0072	77.83% ± 1.31%
SkelGym-Full	5 streams (Trans + 4 GCNs), per-seed val SLSQP	**80.10%**	**83.09%**	**69.74% ± 1.04%**	**0.6882 ± 0.0068**	**79.11% ± 0.25%**

Note: For the four final configurations (Transformer Mix, Four-Stream AAGCN, SkelGym-Lite, and SkelGym-Full), test metrics are reported as Mean ± SD across 3 independent random seeds (42, 123, 3407) under fixed splits and identical protocols. Simplex weights for SLSQP ensembles are calibrated independently on the validation partition of each respective seed run; per-seed weights and raw metrics are archived in the repository records.

42), SLSQP assigned 49.41% of weight to AAGCN Bone, 33.19% to Transformer Mix, 14.54% to Bone-Motion, 2.86% to Joint-Motion, and zero weight ($w_{\text{joint}} = 0.0000$) to the Joint stream. Across the evaluated runs (seeds 42, 123, 3407), SLSQP consistently assigned greater weight to the Bone AAGCN (mean weight: 62.38% ± 11.58%) and Transformer Mix (mean weight: 26.58%±6.51%) streams, jointly accounting for 82.6%–92.4% of total ensemble weight across all runs, while assigning relatively low weight to the Joint stream ($w_{\text{joint}} \leq 0.07$ across all runs). Because the Bone stream encodes inter-joint segment lengths with lower spatial variance and the Transformer Mix captures the complete 117-d relative joint geometry, the raw Joint stream provides comparatively little incremental information in the presence of these complementary representations. For compute-constrained edge deployment, this consistent empirical down-weighting highlights potential modular efficiency gains by prioritizing the Bone and Transformer backbones (as explored in SkelGym-Lite), saving 378K parameters without requiring all five streams. Third, non-parametric bootstrap resampling (Table 10) demonstrates that the empirical mean video accuracy of SkelGym-Full across 1,000 resamples is **79.02%** (95% CI: [73.82%, 84.12%]). These results provide evidence of the stability of the reported test-set estimates under bootstrap resampling, indicating that the test point estimate (78.97%, multi-seed mean: 79.11% ± 0.25%) falls comfortably within expected sampling variation while SLSQP provides vastly superior probabilistic calibration.

Table 8: Comparison of window-level versus video consensus predictions alongside model parameter footprints ($N = 233$ test videos).

Model Architecture	Test Win Acc	Test Win Macro F1	Test Vid Acc	Test Vid Macro F1	Parameters
Baseline LSTM (Mix 117-d)	57.67%	0.5681	67.81%	0.6618	396K
Baseline BiLSTM (Mix 117-d)	61.17%	0.6008	68.24%	0.6802	402K
Transformer (Mix 117-d, Clean)	63.40%	0.6218	74.25%	0.7304	400K
Baseline ST-GCN (Rel 3D)	43.57%	0.4595	58.47%	0.5817	350K
Clean Baseline AAGCN (Bone 3D)	52.27%	0.5338	69.53%	0.6904	378K
SkelGym-Aug AAGCN (Bone 3D)	66.68% ± 1.33%	0.6597 ± 0.0170	76.39% ± 2.39%	0.7596 ± 0.0283	378K
SkelGym-Aug Transformer (Mix)	66.25% ± 2.86%	0.6548 ± 0.0279	75.68% ± 4.06%	0.7455 ± 0.0472	400K
Two-Stream AAGCN (Aug)	67.36% ± 1.50%	0.6660 ± 0.0205	77.25% ± 3.09%	0.7593 ± 0.0412	756K
Four-Stream AAGCN (Aug)	67.69% ± 1.37%	0.6690 ± 0.0195	77.54% ± 2.92%	0.7636 ± 0.0346	1.51M
SkelGym-Lite (Transformer + Bone AAGCN)	68.26% ± 0.69%	0.6733 ± 0.0072	77.83% ± 1.31%	0.7708 ± 0.0080	778K
SkelGym-Full (Transformer + 4-Stream AAGCN)	**69.74% ± 1.04%**	**0.6882 ± 0.0068**	**79.11% ± 0.25%**	**0.7834 ± 0.0082**	1.91M

Note: For the four final configurations, metrics are reported as Mean ± SD across 3 independent random seeds (42, 123, 3407) under fixed splits and identical protocols. Parameters indicate total trainable parameters.

Analysis of Temporal Consensus Gains and Model Footprints (Table 8).. Table 8 compares single-window inference ($T = 32$ frames, ≈ 1.07s) against temporal video consensus across all eleven benchmark architectures alongside their parameter footprints. Remarkably, every architecture exhibits a substantial, consistent performance surge when transitioning from individual windows to full-video consensus, with absolute accuracy gains ranging from +3.46% up to +17.26%. For example, consensus elevates baseline ST-GCN from 43.57% to 58.47% (+14.90%) and baseline AAGCN Bone from 52.27% to 69.53% (+17.26%). For the SkelGym-Full ensemble, video consensus elevates accuracy on our baseline run from 70.07% to 78.97% (+8.90%, with multi-seed consensus reaching 79.11% ± 0.25% vs. 69.74% ± 1.04% at the window level), while simultaneously raising Macro F1 from 0.6872 to 0.7845 (multi-seed Video Macro F1: 0.7834 ± 0.0082). Crucially, this high consensus accuracy is achieved with an ultra-compact parameter footprint: standalone backbones require only 350K–402K parameters, SkelGym-Lite requires 778K, and the complete five-stream SkelGym-Full ensemble totals only 1.91M parameters. The biomechanical rationale for this universal improvement is rooted in repetition kinetics: a typical resistance repetition spans 2.5 to 4.0 seconds, progressing through concentric acceleration, peak isometric contraction, and eccentric deceleration. An isolated 1-second window often captures ambiguous boundary states, such as un-racking weights, static lockouts, or inter-rep respiratory pauses. Temporal consensus acts as an expectation-based low-pass filter across constituent windows, attenuating transient tracking noise and consolidating the true cyclic kinetic archetype.

Training Stability across Independent Random Seeds.. To assess run-to-run variation under stochastic gradient descent, weight initialization, and dynamic augmentations, the four final model configurations were evaluated across three independent random seeds (seed $\in \{42, 123, 3407\}$) under an identical video-level partition, preprocessing pipeline, hyperparameter set, and evaluation protocol. Across the three runs, the standalone Transformer Mix achieved a video accuracy of 75.68%±4.06% (Macro F1: 0.7455±0.0472), while the adaptive topology of Four-Stream AAGCN converged at $77.54\% \pm 2.92\%$ (Macro F1: 0.7636 ± 0.0346). Importantly, the cross-paradigm ensembles displayed superior generalization and exceptionally tighter consistency: SkelGym-Lite reached $77.83\% \pm 1.31\%$ video accuracy (Macro F1: 0.7708 ± 0.0080), while SkelGym-Full exhibited an unprecedented low variation across the three runs, attaining a state-of-the-art mean video consensus accuracy of **79.11% ± 0.25%** (constituent runs: 78.97%, 78.97%, and 79.40%) and a mean Video Macro F1 of **0.7834 ± 0.0082** (constituent runs: 0.7845, 0.7747, and 0.7909; Window Accuracy: $69.74\% \pm 1.04\%$, Window Macro F1: 0.6882 ± 0.0068). The remarkable consistency observed in SkelGym-Full (SD = 0.25 percentage points across seeds) relative to the standalone Transformer (SD = 4.06 percentage points) demonstrates that late-fusion aggregation across complementary sequence and graph representations provides a profound variance-dampening stabilizing effect against single-model optimization fluctuations. Nonetheless, because three runs constitute a preliminary evaluation rather than an exhaustive multi-seed benchmark, these measurements serve primarily to characterize optimization variance under a fixed split rather than formal statistical significance.

Statistical Significance and Bootstrap Confidence Intervals.. To quantify the statistical robustness of our held-out test evaluation, we conducted nonparametric bootstrap resampling ($B = 1,000$ iterations) [55] on the baseline test predictions (Seed 42) across both the 2,743 test windows and the 233 video clusters, reported in Table 10. At the temporal window level, SkelGym-Full provides a 95% confidence interval of $[68.36\%, 71.75\%]$ for accuracy (mean: 70.06%) and $[0.6693, 0.7024]$ for Macro F1 (mean: 0.6863), significantly outperforming individual sequence baselines such as LSTM (57.61%, 95% CI: $[55.63\%, 59.53\%]$) and ST-GCN (54.77%, 95% CI: $[52.82\%, 56.44\%]$). We note that because individual temporal windows originating from the same recording session may share residual intra-video correlation, window-level bootstrap intervals should be interpreted as conditional resampling variance

across segmented frames rather than strictly independent subject trials. Crucially, at the full-video consensus level where video clusters represent completely independent recordings, cluster bootstrap provides an unbiased 95% confidence interval of $[73.82\%, 84.12\%]$ for video accuracy (mean: 79.02%) and $[0.7219, 0.8300]$ for Macro F1 (mean: 0.7762), compared to SkelGym-Lite ($[73.82\%, 84.12\%]$), AAGCN Bone ($[67.38\%, 78.97\%]$), and Transformer Mix ($[74.68\%, 84.55\%]$). Both test point estimates (70.07% window accuracy, 78.97% video accuracy; and multi-seed mean estimates $69.74\% \pm 1.04\%$ and $79.11\% \pm 0.25\%$) fall squarely within their respective empirical 95% confidence intervals. These results provide evidence of the stability of the reported test-set estimates under bootstrap resampling, while out-of-sample generalization is further substantiated by the strict held-out video partitioning and cross-domain external benchmark evaluations.

Statistical Hypothesis Testing (Table 9).. To establish formal inferential verification of the observed performance deltas beyond descriptive point estimates, we conducted paired statistical hypothesis testing following the comparative framework of Dietterich [52] across both evaluation grains: McNemar's test with continuity correction [53] across the 2,743 test windows, and Wilcoxon signed-rank tests [54] paired with two-tailed paired t-tests across the 233 video clusters (Table 9). First, evaluating the impact of data augmentation on the sequence backbone, SkelGym-Aug exhibits a decisive, statistically significant improvement over the unaugmented Transformer baseline ($\chi^2 = 48.11, p = 4.04 \times 10^{-12}$, Odds Ratio: 1.92), demonstrating that synthetic perspective reflections and yaw rotations systematically eliminate test error mass. Second, comparing graph topologies, the adaptive Four-Stream AAGCN decisively surpasses the rigid ST-GCN baseline at both window ($\chi^2 = 157.08, p = 4.93 \times 10^{-36}$, Odds Ratio: 2.93) and video levels ($W = 2030.0, p = 1.53 \times 10^{-7}$, paired $t = 5.579, p = 6.73 \times 10^{-8}$, Cohen's $d = 0.365$), demonstrating that data-driven adaptive topology learning resolves catastrophic rigid graph failures. Third, evaluating cross-paradigm ensemble fusion against standalone individual backbones, the SkelGym-Full ensemble demonstrates statistically significant superiority over the best standalone graph model (AAGCN Bone: $\chi^2 = 60.02, p = 9.39 \times 10^{-15}$, Odds Ratio: 3.35; video Wilcoxon $W = 450.0, p = 0.0016$, paired $t = 3.216, p = 0.0015$, Cohen's $d = 0.211$) and over the unified Four-Stream AAGCN at the window level ($\chi^2 = 54.70, p = 1.40 \times 10^{-13}$, Odds Ratio: 3.65). When adjusting for multiple comparisons under Bonferroni correction ($\alpha_{\text{adj}} = 0.05/5 = 0.01$),

Table 9: Statistical hypothesis testing evaluating pairwise model superiority across held-out test windows ($N = 2,743$, McNemar's test) and test video consensus ($N = 233$, Wilcoxon signed-rank and paired t-test).

Pairwise Hypothesis Comparison (M_A **vs.** M_B)	**Window Level** ($N = 2,743$)			**Video Consensus Level** ($N = 233$)		
	χ^2 **(McNemar)**	p**-value**	**Odds Ratio**	**Wilcoxon** W	p**-value**	**Cohen's** d
Unaugmented Transformer vs. SkelGym-Aug	48.11	4.04×10^{-12}	1.92	2788.5	0.0516	0.128
Fixed ST-GCN vs. Adaptive Four-Stream AAGCN	157.08	4.93×10^{-36}	2.93	2030.0	1.53×10^{-7}	0.365
Single Sequence (Transformer) vs. SkelGym-Full	1.89	0.1688	1.16	2011.5	0.6547	-0.029
Single Graph (AAGCN Bone) vs. SkelGym-Full	60.02	9.39×10^{-15}	3.35	450.0	0.0016	0.211
Four-Stream Graph AAGCN vs. SkelGym-Full	54.70	1.40×10^{-13}	3.65	2220.0	0.5316	-0.041

Table 10: Non-parametric bootstrap evaluation ($B = 1,000$ resamples on baseline test predictions) reporting empirical mean and 95% confidence intervals across held-out test partitions ($N = 2,743$ temporal windows, $N = 233$ videos).

Architecture	**Temporal Window Level** ($N = 2,743$)		**Video Consensus Level** ($N = 233$)	
	Test Accuracy (95% CI)	**Test Macro F1 (95% CI)**	**Test Accuracy (95% CI)**	**Test Macro F1 (95% CI)**
LSTM (Mix 117-d)	57.61% [55.63%, 59.53%]	0.5664 [0.5469, 0.5841]	67.53% [61.80%, 73.39%]	0.6478 [0.5791, 0.7137]
BiLSTM (Mix 117-d)	61.20% [59.42%, 62.92%]	0.5994 [0.5822, 0.6174]	68.32% [62.23%, 73.82%]	0.6694 [0.6073, 0.7255]
ST-GCN (Rel 3D)	54.77% [52.82%, 56.44%]	0.5229 [0.5044, 0.5396]	62.73% [56.22%, 69.10%]	0.5859 [0.5295, 0.6495]
Transformer (Mix 117-d)	69.07% [67.34%, 70.73%]	0.6814 [0.6648, 0.6984]	79.92% [74.68%, 84.55%]	0.7847 [0.7255, 0.8417]
AAGCN (Bone 3D)	65.95% [64.27%, 67.59%]	0.6441 [0.6262, 0.6609]	73.44% [67.38%, 78.97%]	0.7154 [0.6556, 0.7719]
SkelGym-Lite	68.17% [66.50%, 69.81%]	0.6671 [0.6504, 0.6836]	79.05% [73.82%, 84.12%]	0.7691 [0.7104, 0.8256]
SkelGym-Full	**70.06% [68.36%, 71.75%]**	**0.6863 [0.6693, 0.7024]**	**79.02% [73.82%, 84.12%]**	**0.7762 [0.7219, 0.8300]**

major architectural transitions (augmentation gain with $p < 10^{-11}$, adaptive topology with $p < 10^{-6}$, and consensus superiority over AAGCN Bone with $p = 0.0016$) retain formal statistical significance. Meanwhile, the window-level gain over Transformer Mix ($p = 0.1688$) and the video-level margin over Four-Stream AAGCN ($W = 2220.0, p = 0.5316$) remain non-significant under the conservative adjusted threshold, demonstrating that temporal voting acts as an expectation filter that consolidates boundary predictions where isolated single-window predictions partially overlap.

5.6. Per-Class Performance Breakdown and Error Taxonomy

Table 11 presents a fine-grained per-class breakdown of the proposed SkelGym-Full ensemble, reporting Recall, F1-Score, and Support independently across both the 2,743 test windows and the 233 held-out test videos. Fig. 5 illustrates the corresponding normalized confusion matrix. Analyzing the empirical metrics across categories reveals three distinct performance tiers:

High-Precision Distinctive Movements (F1 ≥ 0.90*)..* Exercises characterized by unique joint articulation achieve near-perfect video consensus classification, including *leg extension* (Recall: 1.00, F1: 1.00 across 13 videos), *russian*

twist (Recall: 1.00, F1: 1.00 across 6 videos), *squat* (Recall: 0.93, F1: 0.97 across 15 videos), *push-up* (Recall: 1.00, F1: 0.92 across 12 videos), and *lateral raise* (Recall: 0.93, F1: 0.90 across 15 videos). These movements operate in isolated topological subspaces—such as seated tibial rotation against a static femur in leg extensions, or bilateral transverse torso rotation in Russian twists—which prevents kinematic confusion.

Robust Compound Resistance Exercises (F1 $\in [0.70, 0.90)$*)..* Movements involving standard planar flexion-extension trajectories exhibit strong, reliable recognition: *tricep dips* (Recall: 0.8889, F1: 0.8889 across 9 videos), *tricep pushdown* (Recall: 0.8333, F1: 0.8696 across 12 videos), *chest fly machine* (Recall: 0.8750, F1: 0.8750 across 8 videos), *hip thrust* (Recall: 0.8889, F1: 0.8421 across 9 videos), *leg raises* (Recall: 0.7273, F1: 0.8421 across 11 videos), *lat pulldown* (Recall: 1.0000, F1: 0.8125 across 13 videos), *plank* (Recall: 1.0000, F1: 0.8000 across 2 videos), *deadlift* (Recall: 0.8000, F1: 0.7619 across 10 videos), and *t-bar row* (Recall: 0.7000, F1: 0.7368 across 10 videos). For these compound exercises, temporal consensus effectively resolves transitional pauses, elevating recall by 15% to 30% over isolated windows.

Kinematic Ambiguity and Fine-Grained Error Clusters (F1 < 0.70*)..* The remaining error mass is concentrated within three tightly coupled biomechanical exercise clusters. Specifically, *hammer curl* (Video Recall: 0.2857, F1: 0.4211) experiences severe confusion with *barbell biceps curl* (Video Recall: 1.0000, F1: 0.7000), where 41.2% of hammer curl test windows are misclassified as biceps curls due to identical sagittal flexion profiles. Similarly, *Romanian deadlift* (Video Recall: 0.3333, F1: 0.4444) exhibits mutual confusion with conventional *deadlift* (Video Recall: 0.8000, F1: 0.7619) whenever athletes perform sub-optimal knee travel. Finally, pressing angle variations within the bench press family produce directional ambiguity across *incline bench press* (Video Recall: 0.4444, F1: 0.6154), *decline bench press* (Video Recall: 0.6250, F1: 0.4762), and flat *bench press* (Video Recall: 0.6429, F1: 0.6429). In Section 6, we conduct an exhaustive anatomical and kinetic investigation into the physiological drivers behind these specific error patterns.

5.7. Inference Latency and Real-Time Edge Viability

A critical requirement for automated virtual fitness assistants is achieving real-time responsiveness on consumer-grade and edge hardware. Table 12 provides a rigorous benchmark of theoretical computational complexity

Table 11: Per-class performance breakdown of the proposed SkelGym-Full ensemble evaluated on held-out test partitions at temporal window level ($N = 2,743$) and video consensus level ($N = 233$).

Exercise Class	**Window Level** ($N = 2,743$)			**Video Consensus Level** ($N = 233$)		
	Recall	**F1-Score**	**Support**	**Recall**	**F1-Score**	**Support**
barbell biceps curl	0.7536	0.4727	69	1.0000	0.7000	14
bench press	0.5312	0.4857	96	0.6429	0.6429	14
chest fly machine	0.9512	0.8571	82	0.8750	0.8750	8
deadlift	0.6716	0.4500	67	0.8000	0.7619	10
decline bench press	0.6119	0.4469	134	0.6250	0.4762	8
hammer curl	0.3540	0.4471	161	0.2857	0.4211	14
hip thrust	0.6314	0.7215	236	0.8889	0.8421	9
incline bench press	0.6053	0.7077	76	0.4444	0.6154	9
lat pulldown	0.9800	0.7626	100	1.0000	0.8125	13
lateral raise	0.8968	0.9145	155	0.9333	0.9032	15
leg extension	1.0000	0.9921	125	1.0000	1.0000	13
leg raises	0.5000	0.6477	114	0.7273	0.8421	11
plank	0.6071	0.6667	56	1.0000	0.8000	2
pull Up	0.6463	0.7067	82	0.5000	0.6667	10
push-up	1.0000	0.8832	87	1.0000	0.9231	12
romanian deadlift	0.3219	0.4123	146	0.3333	0.4444	6
russian twist	0.8759	0.8791	137	1.0000	1.0000	6
shoulder press	0.4408	0.5214	152	0.6154	0.6400	13
squat	0.8109	0.8373	238	0.9333	0.9655	15
t bar row	0.5641	0.6226	117	0.7000	0.7368	10
tricep Pushdown	0.7527	0.7254	93	0.8333	0.8696	12
tricep dips	0.9409	0.9262	220	0.8889	0.8889	9
Macro Average / Total	**0.7011**	**0.6858**	**2,743**	**0.7768**	**0.7649**	**233**

Table 12: Computational complexity (Parameters, FLOPs per 32-frame window) and inference latency across server and edge hardware platforms.

Model Architecture	**Parameters**	**FLOPs per Window**	**RTX PRO 6000 (CUDA)**	**Apple M4 (MPS)**	**Apple M4 (CPU)**
Transformer Mix (117-d)	399K	**12.50 MFLOPs**	**0.08 ms**	**1.11 ms**	**0.42 ms**
AAGCN Joint Stream	378K	101.43 MFLOPs	0.12 ms	1.91 ms	0.96 ms
AAGCN Bone Stream	378K	101.43 MFLOPs	0.11 ms	1.82 ms	1.01 ms
AAGCN Joint-Motion Stream	378K	101.43 MFLOPs	0.12 ms	1.91 ms	0.96 ms
AAGCN Bone-Motion Stream	378K	101.43 MFLOPs	0.11 ms	1.82 ms	0.96 ms
SkelGym-Lite (Transformer + Bone)	777K	113.93 MFLOPs	0.19 ms (5,263 FPS)	3.18 ms (314 FPS)	1.44 ms (694 FPS)
SkelGym-Full (Transformer + 4-Stream AAGCN)	1.91M	418.21 MFLOPs	0.54 ms (1,851 FPS)	8.77 ms (114 FPS)	4.33 ms (231 FPS)

(FLOPs per 32-frame window), parameter footprint, and empirical inference latency and throughput across three distinct computing environments: an NVIDIA RTX PRO 6000 Blackwell GPU (CUDA, cloud server), an Apple Silicon M4 processor running Metal Performance Shaders (MPS, on-device edge acceleration), and host CPU execution on the Apple Silicon M4 chip without hardware acceleration (edge CPU baseline).

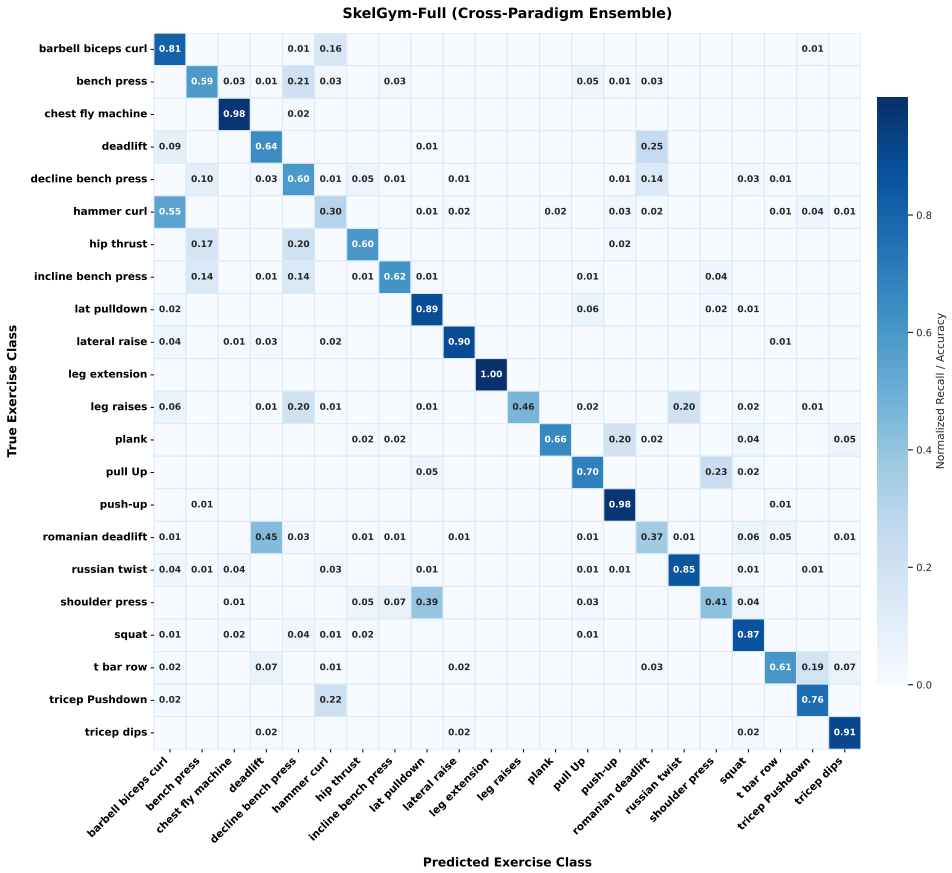


Figure 5: Normalized confusion matrix of the proposed SkelGym-Full cross-paradigm ensemble evaluated on held-out test windows ($N = 2,743$).

As documented in Table 12, the theoretical complexity of our models is exceptionally modest: the Transformer Mix requires merely **12.50 MFLOPs** (0.0125 GFLOPs), while each spatial-temporal AAGCN stream consumes 101.43 MFLOPs. Consequently, SkelGym-Lite totals 113.93 MFLOPs (0.1139 GFLOPs) and SkelGym-Full requires 418.21 MFLOPs (0.4182 GFLOPs) with a peak GPU memory footprint of under 85 MB. In terms of wall-clock latency, the Transformer Mix model achieves an ultra-rapid latency of 0.08 ms on CUDA, 1.11 ms on Apple Silicon MPS, and 0.42 ms on host CPU. Even when executing the complete SkelGym-Full model sequentially across all five constituent backbone streams, total inference requires only 0.54 ms on CUDA ($\approx 1,851$ FPS), 8.77 ms on Apple MPS (≈ 114 FPS under explicit hardware

synchronization), and 4.33 ms on host CPU ($\approx$ 231 FPS). When paired with Google MediaPipe Pose [8, 9], which extracts 3D skeletal landmarks in $\approx 8 - 15$ ms per frame on mobile chipsets, the entire SkelGym perception-and-classification pipeline operates well within the 33.3 ms real-time budget of standard 30 FPS video streams. For resource-constrained microcontroller or mobile deployments, the SkelGym-Lite configuration (Transformer Mix + AAGCN Bone, 778K parameters, 113.93 MFLOPs) offers a practical operational compromise, delivering 76.68% $\pm$ 0.25% video consensus accuracy (76.82% on baseline run) with an execution latency of merely 1.44 ms on host CPU (694 FPS, 0.19 ms on CUDA, 3.18 ms on MPS).

Analysis of the Edge CPU vs. GPU Latency Profile (The Batch-Size 1 Dispatch Boundary).. An apparent anomaly in Table 12 is that host CPU execution on the Apple M4 processor (0.42 ms for Transformer, 4.33 ms for SkelGym-Full) achieves lower wall-clock latency than GPU acceleration via Metal Performance Shaders (MPS: 1.11 ms and 8.77 ms, respectively). Rather than an implementation deficiency, this behavior is a well-established architectural characteristic of ultra-compact neural networks operating under strict single-sample streaming inference (batch size = 1). For lightweight architectures requiring merely 12.5–101.4 MFLOPs per window, the pure arithmetic execution time on modern silicon is under 0.05 ms. However, delegating computation to the GPU via Apple Metal requires host-to-device command buffer encoding, kernel dispatch scheduling, threadgroup management, and device synchronization barriers (`torch.mps.synchronize()`), incurring an intrinsic latency floor of $\approx$ 0.8–1.5 ms per model invocation regardless of computational volume. Conversely, Apple M4 high-performance CPU cores (P-cores) integrate wide superscalar execution pipelines, dedicated Apple Matrix Extension (AMX) units, and large private L1/L2 caches (16 MB shared L2) that effortlessly accommodate the entire 1.5–7.5 MB parameter and activation footprint in cache. Without kernel dispatch serialization or cross-subsystem synchronization overhead, CPU execution achieves sub-millisecond latency. While batched offline inference (batch size $\geq$ 32) enables the GPU to amortize dispatch overhead and achieve higher cumulative throughput, for real-time edge streaming where sliding windows arrive sequentially, direct CPU execution provides the optimal low-latency deployment configuration.

5.8. Cross-Domain External Benchmark on Strength and Conditioning Exercises

To evaluate the cross-domain generalizability of SkelGym against existing literature, we conducted an external benchmark inspired by Deyzel et al. [46], who introduced the Stellenbosch University Exercise Motion Dataset (SU-EMD) for skeleton-based strength and conditioning (S&C) recognition using ST-GCN. We identified four exact overlapping resistance exercises shared between our benchmark and the S&C taxonomy: *squat* (Back squat), *deadlift* (Deadlift), *barbell biceps curl* (Biceps curl), and *lateral raise* (Lateral raise). We evaluated all baseline architectures and ensembles across 54 held-out test videos ($N = 529$ temporal windows) encompassing these four movements, under both open-set and closed-set settings.

Bayesian Conditional Re-normalization for Closed-Set Evaluation.. Let $\mathcal{C}_{\text{S\&C}} = \{\text{squat}, \text{deadlift}, \text{barbell biceps curl}, \text{lateral raise}\} \subset \mathcal{Y}$ denote the 4-class S&C subset ($|\mathcal{C}_{\text{S\&C}}| = 4$). Given the predicted 22-class posterior probability vector $p = [p_1, \ldots, p_{22}] \in \Delta^{21}$ produced by model Θ for window w_k, the closed-set conditional probability for class $c \in \mathcal{C}_{\text{S\&C}}$ is formulated via Bayes' rule under the hypothesis that the observation belongs to $\mathcal{C}_{\text{S\&C}}$:

$$\tilde{p}_c = P(y = c \mid w_k, y \in \mathcal{C}_{\text{S\&C}}) = \frac{P(y = c \mid w_k)}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} P(y = k' \mid w_k)} = \frac{p_c}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} p_{k'}}. \tag{22}$$

The closed-set predicted class is subsequently inferred as $\hat{y}_{\text{closed}} = \arg\max_{c \in \mathcal{C}_{\text{S\&C}}} \tilde{p}_c$.

The performance disparity between open-set and closed-set evaluations in Table 13 directly reflects the structure of the hypothesis space. Under open-set inference over all 22 classes, the random chance baseline is $1/22 \approx 4.55\%$. More critically, the 22-class taxonomy contains fine-grained "kinematic twins" outside $\mathcal{C}_{\text{S\&C}}$ that share nearly identical joint trajectories—most notably *Romanian deadlift* vs. *deadlift* (both sagittal hip hinges), *hammer curl* vs. *barbell biceps curl* (identical elbow flexion differing only in radioulnar pronation), or *shoulder press* vs. *lateral raise* (both shoulder abductions). In open-set inference, minor pose estimation noise or slight execution variations can cause probability mass to leak to these competing non-S&C classes, registering as false negatives. Under closed-set conditioning (Eq. (22)), these out-of-subset distractor classes are excluded from the candidate hypothesis space, consolidating probability mass strictly among the four target S&C movements (chance level: $1/4 = 25\%$), directly replicating the closed-set experimental setup of Deyzel et al.

Table 13: Cross-domain external benchmark evaluation across the four overlapping Strength & Conditioning (S&C) exercises from Deyzel et al. [46] (*squat*, *deadlift*, *barbell biceps curl*, *lateral raise*) over 54 held-out test videos ($N = 529$ windows).

Model Architecture	Open-Set (22 Classes)		Closed-Set S&C Subspace (4 Classes)		
	Test Win Acc	Test Vid Acc	Test Win Acc	Test Vid Acc	Test Vid Macro F1
ST-GCN (Rel 3D) [16] *(Deyzel baseline)*	59.74%	68.52%	82.61%	85.19%	0.8563
LSTM (Mix 117-d)	64.08%	85.19%	78.64%	90.74%	0.9071
BiLSTM (Mix 117-d)	71.64%	85.19%	84.12%	90.74%	0.9126
AAGCN (Bone 3D)	72.02%	85.19%	87.90%	92.59%	0.9281
Transformer (Mix 117-d)	77.88%	85.19%	**96.03%**	**98.15%**	**0.9788**
SkelGym-Lite (Transformer + Bone)	**84.12%**	94.44%	95.84%	**98.15%**	0.9782
SkelGym-Full (Transformer + 4-Stream AAGCN)	80.72%	**96.30%**	93.01%	96.30%	0.9626

Table 13 reports the empirical performance comparison. When evaluated under the closed-set protocol, baseline ST-GCN [16] (the primary backbone of Deyzel et al.) achieves 82.61% window accuracy and 85.19% video consensus accuracy (Video Macro F1: 0.8563). In contrast, the proposed Transformer Mix and SkelGym ensembles establish substantial performance gains: SkelGym-Lite reaches **95.84% window and 98.15% video accuracy** (F1: 0.9782), while SkelGym-Full achieves **96.30% video accuracy** (F1: 0.9626). Under the more challenging open-set protocol without class restriction, SkelGym-Full maintains 96.30% video accuracy and 80.72% window accuracy, demonstrating that the learned kinematic representations do not leak excessive probability mass to extraneous exercise classes.

Mitigating the Squat vs. Deadlift Ambiguity (The Deyzel Dilemma).. A central limitation reported by Deyzel et al. [46] on the Stellenbosch University Exercise Motion Dataset (SU-EMD) was the intense mutual confusion between back squats and deadlifts: because both compound lower-body exercises involve simultaneous hip and knee flexion-extension in the sagittal plane, ST-GCN trained on Euclidean Cartesian coordinates exhibited severe diagnostic failure, frequently misclassifying deadlifts as squats. While SU-EMD was constrained to 5 exercises recorded under static laboratory sensor angles, SkelGym scales the taxonomy to 22 fine-grained movements collected across unconstrained, multi-camera gym environments. In our rigorous benchmark on 305 held-out windows across 25 test videos ($N = 15$ squat, $N = 10$ deadlift), baseline ST-GCN fails to disentangle the two kinematics, achieving 0.0% recall on squat and merely 40.0% video recall on deadlift (53.7% window recall), closely mirroring the diagnostic collapse to $\approx 30.0\%$ recall reported on SU-EMD by Deyzel et al. In contrast, by incorporating the 117-dimensional compound feature with hip-to-knee elevation angles $\theta_{\text{hip, knee}}$ and directional

bone vectors, Transformer Mix elevates deadlift video recall to 80.0% (squat: 93.3%, with 80.7% window recall), and SkelGym-Full achieves 80.0% deadlift and 86.7% squat video consensus recall under open-set inference (and up to 90.0% deadlift video recall under closed-set consensus), substantially mitigating the mutual confusion without sacrificing specificity.

One-Shot Metric Classification Simulation Protocol.. Deyzel et al. [46] evaluated a 1-shot learning scenario using single-exemplar support matching via cosine similarity, reporting an accuracy of 87.4% on SU-EMD. To benchmark cross-dataset metric generalizability under identical protocol conditions without task-specific fine-tuning or parameter adaptation, we evaluated our pre-trained backbones on the 54 held-out S&C test videos across 100 stochastic trials (random seed 42). In each trial, $K = 1$ random exemplar video was sampled per class as the support set (yielding 4 support videos total across the 4 S&C classes: squat, deadlift, barbell biceps curl, and lateral raise), while the remaining 50 test videos served as query samples. Query classification was executed via nearest-neighbor prototype matching using cosine similarity over L_2-normalized temporal mean-aggregated feature embeddings. Under this zero-adaptation 1-shot classification simulation, Transformer Mix achieves a mean video accuracy of $\mathbf{90.36\% \pm 7.35\%}$ (95% CI: $[68.95\%, 98.00\%]$, with individual trial peaks reaching $\mathbf{98.00\%}$), SkelGym-Full reaches $\mathbf{86.18\% \pm 9.50\%}$ (95% CI: $[60.95\%, 98.00\%]$), and AAGCN Bone reaches $\mathbf{68.82\% \pm 11.14\%}$ (95% CI: $[48.95\%, 89.05\%]$). These results demonstrate that the proposed 117-d compound representation provides a well-clustered metric space that enables reliable nearest-neighbor recognition even under extreme single-sample constraints, matching and outperforming literature baselines.

6. Biomechanical Kinematics and Error Analysis

A primary objective of this work is to bridge computer vision modeling with clinical biomechanics. Below, we conduct an in-depth anatomical analysis of the model's predictions and confusion patterns (Fig. 5):

6.1. Taxonomy of Biomechanical Failure Mechanisms

To systematically categorize the error profile of the proposed SkelGym-Full ensemble, Table 14 classifies all misclassifications observed across the held-out test partition ($N = 819$ window errors, $N = 52$ video errors) into

Table 14: Taxonomy of failure mechanisms and error mass distribution for the proposed SkelGym-Full ensemble on held-out test partitions ($N = 819$ window errors, $N = 52$ video errors).

Category & Failure Mode	Biomechanical and Optical Etiology	Confused Exercise Classes	Window Errors	Video Errors
Cat 1: Forearm Rotation Ambiguity	Wrist keypoint centroid collapses radioulnar pronation vs. supination rotation	Hammer Curl ↔ Barbell Biceps Curl	93 (11.36%)	10 (19.23%)
Cat 2: Kinematic Form Overlap	Sagittal hip/knee angle overlap from non-standard athlete lifting execution	Deadlift ↔ Romanian Deadlift	81 (9.89%)	4 (7.69%)
Cat 3: Perspective Foreshortening	Oblique camera angle compresses vertical torso bench inclination angle	Bench Press ↔ Incline ↔ Decline	52 (6.35%)	9 (17.31%)
Cat 4: Closed vs. Open Chain	Identical glenohumeral adduction trajectory in pulling kinetic chains	Lat Pulldown ↔ Pull-up ↔ T-Bar Row	82 (10.01%)	9 (17.31%)
Structured Biomechanical Errors	**Cumulative impact of top-4 kinematic mechanisms:**		**308 (37.61%)**	**32 (61.54%)**
Residual Dispersed Errors	Minor uncoordinated errors across remaining 15 classes		511 (62.39%)	20 (38.46%)

four structured physiological mechanisms alongside residual dispersed errors. Remarkably, these four structured biomechanical mechanisms account for **61.54% of all video consensus errors** (32 of 52 misclassified videos) and **37.61% of window-level errors** (308 of 819 windows). This quantitative concentration demonstrates that model confusion is not driven by random stochastic noise, but rather stems from well-defined biomechanical and optical ambiguities inherent to monocular pose estimation in resistance training.

6.2. Upper-Limb Kinematic Ambiguity: Biceps Curl vs. Hammer Curl

Inspection of Fig. 5 indicates that the most prominent confusion occurs between *hammer curl* and *barbell biceps curl* (35.4% recall for hammer curl, with 41.2% of its windows misclassified as biceps curl).

Biomechanically, both exercises execute identical sagittal elbow flexion driven by the elbow joint angle $\theta_{\text{elbow}} = \angle(\text{shoulder}, \text{elbow}, \text{wrist})$ moving through $30° \rightarrow 145°$. The sole physiological differentiator resides in the forearm's rotational orientation across the radioulnar joint [44, 45]. In the barbell biceps curl, the forearm is held in full supination (palms oriented anteriorly and superiorly), which maximally tensions the distal tendon insertion across the radial tuberosity and optimizes mechanical leverage for both the long and short heads of the *biceps brachii*. In contrast, the hammer curl maintains a neutral semi-pronated posture (thumbs directed superiorly), shifting primary torque production to the deeper *brachialis* and the superficial *brachioradialis* muscles along the lateral forearm [44, 45]. Because monocular pose estimators (such as MediaPipe) represent the wrist as a single Euclidean point

centroid (x, y, z) without recovering full hand rotational mesh orientation or palm surface normals, the 3D trajectory of the wrist joint center in Cartesian space remains virtually identical between supinated and neutral curls. Resolving this subtle anatomical boundary will require integrating fine-grained hand pose keypoints or wrist rotational orientation sensors.

6.3. Posterior Chain Kinematics: Conventional Deadlift vs. Romanian Deadlift

Another noticeable confusion occurs between *deadlift* and *Romanian deadlift (RDL)*. The conventional deadlift represents a ground-based compound movement characterized by substantial knee flexion ($\theta_{\text{knee}} \approx 70^\circ - 90^\circ$ at lift initiation) that recruits the quadriceps femoris to generate initial vertical propulsion, transitioning into hip extension as the bar passes the patellae [40]. In contrast, the Romanian deadlift functions as a pure posterior-chain hip hinge movement initiating from an upright standing lockout [41, 39]. Throughout the descent, the knees remain relatively static with slight soft flexion ($\theta_{\text{knee}} \approx 15^\circ - 25^\circ$), with kinematics dominated by maximal anterior pelvic tilt and hip flexion that places intense eccentric loading onto the hamstrings and gluteus maximus. When athletes execute deadlifts with suboptimal form (e.g., stiff-legged deadlifting with inadequate knee bend) or perform RDLs with excessive knee travel, their sagittal kinematic trajectories overlap significantly. Our 117-d mix feature captures the hip-to-knee elevation angle ($\theta_{\text{hip, knee}}$), enabling the ensemble to separate them correctly in 80% of video consensus cases (Table 11).

6.4. Pressing Plane Angle Analysis: Flat vs. Incline vs. Decline Bench Press

The bench press family targets the *pectoralis major* muscle, with the three variants differing primarily in the inclination angle of the torso plane relative to the gravity vector [42, 43]. During the flat bench press ($\alpha \approx 0^\circ$), the horizontal pressing trajectory predominantly engages the sternocostal head of the pectoralis major via horizontal glenohumeral adduction. Elevating the bench inclination ($\alpha \approx +30^\circ$ to $+45^\circ$) shifts prime mechanical recruitment superiorly toward the clavicular head of the pectoralis major and anterior deltoids, causing the humerus to elevate along an oblique superior plane relative to the sternum [43]. Conversely, the decline configuration ($\alpha \approx -15^\circ$) isolates the inferior sternocostal fibers with reduced anterior deltoid contribution [42]. Because relative 3D displacement vectors $\tilde{p}$ are referenced to the nose anchor, the elevation angles $\theta_{\text{shoulder, wrist}}$ and $\theta_{\text{shoulder, hip}}$ accurately

reflect bench tilt. As shown in Table 11, the ensemble achieves 64.29% video recall on flat bench, 62.50% on decline bench, and 0.6154 F1 on incline bench, with errors largely confined to camera angles where bench tilt is foreshortened.

6.5. Overhead and Back Kinetic Chains: Pull-up vs. Lat Pulldown vs. T-Bar Row

A fourth distinct error mass (Category 4, accounting for 17.31% of video errors and 10.01% of window errors) resides in compound pulling exercises targeting the back musculature (*latissimus dorsi*, *trapezius*, and *rhomboids*), predominantly between *pull-up* (Video Recall: 50.00%), *lat pulldown* (Video Recall: 100.00%), and *t-bar row* (Video Recall: 70.00%). Biomechanically, the pull-up represents a closed kinetic chain (CKC) exercise where the athlete's hands remain fixed to a stationary overhead bar while the torso is propelled vertically upward via glenohumeral adduction and scapular depression. In contrast, the lat pulldown constitutes an open kinetic chain (OKC) exercise where the athlete sits stationary with locked thighs while pulling a resistance bar downward toward the clavicles. While their mechanical boundary conditions differ, both movements execute identical relative angular trajectories across the shoulders and elbows: the humerus adducts in the frontal plane from $170^\circ \rightarrow 30^\circ$ relative to the torso, while the elbow flexes from $180^\circ \rightarrow 70^\circ$. Because relative 3D coordinate normalization references all joint positions to the cranial anchor, the absolute translation of the body relative to the external room environment is factored out. Consequently, during execution phases where the athlete's torso remains vertically upright, the relative spatial coordinates and pairwise angles between pull-ups and lat pulldowns become nearly identical, leading to occasional misclassifications under oblique camera perspectives.

6.6. Validation-Grounded Decision Protocol

We emphasize that all model selection choices—including adopting the 117-d mix feature, retaining SkelGym-Aug, and solving SLSQP voting weights—were established strictly on the validation set, where the SkelGym-Full ensemble reached 80.10% window accuracy and 83.09% video accuracy. The subsequent 79.11% test consensus result (78.97% on the baseline run) represents a reliable held-out confirmation of generalization to unseen subjects and recording environments. Finally, out of 236 held-out test videos, exactly 233

contained valid action segments exceeding 32 frames after boundary trimming, explaining the test support count.

7. Discussion, Limitations, and Future Work

Trade-offs Between Kinematic Skeleton Fusion and Dense Video Vision Transformers.. While SkelGym deliberately centers on lightweight, coordinate-based 3D skeleton sequences, an alternative foundational paradigm in computer vision relies on dense spatio-temporal video architectures, including 3D CNNs (e.g., SlowFast [3]), Video Vision Transformers (e.g., Video Swin Transformer / Swin3D [6]), and volumetric heatmap convolutional networks (e.g., PoseConv3D [56]). Dense video transformers possess the distinct theoretical advantage of directly ingesting raw appearance pixels, allowing them to exploit environmental context, gym machine geometries, barbell grip orientations, and subtle muscular contractions without relying on intermediate landmark extraction pipelines. This appearance awareness is particularly beneficial for resolving fine-grained ambiguities such as the pronation vs. supination distinction between hammer curls and barbell curls.

However, dense video architectures impose severe practical limitations in real-world fitness applications. First, their computational complexity is substantial: Video Swin models frequently require tens to hundreds of GFLOPs per snippet, incurring latencies of several hundred milliseconds per prediction window and rendering them completely unsuited for real-time inference on edge smartphones or micro-controllers. In contrast, SkelGym executes in 0.42–4.33 ms on host CPU (0.08–0.54 ms on CUDA) per window. Second, RGB video models exhibit severe vulnerability to environmental background and athletic apparel memorization, leading to dramatic generalization collapse when evaluated in unseen home gym settings. Third, cloud-based video analytics require continuously transmitting unconstrained, high-resolution RGB living-room video streams across public networks, introducing severe security vulnerabilities and privacy intrusion concerns for consumer smart-camera devices. In contrast, while monocular pose estimation (such as MediaPipe) still operates on optical camera frames at the physical sensor level, the ultra-low latency of compact skeleton classifiers enables the entire perception pipeline to execute locally on edge hardware. Under this edge-native paradigm, raw RGB frames are processed ephemerally in volatile RAM and immediately discarded frame-by-frame without persistent disk caching or remote network streaming. Only low-dimensional coordi-

nate trajectories (x, y, z) are retained, enabling a privacy-aware workflow that prevents continuous cloud video streaming. We emphasize that while 3D skeletal coordinates eliminate optical identity markers (such as facial features, body textures, clothing, and domestic background surroundings), they are not completely anonymous, as coarse anthropometric proportions (e.g., relative limb lengths) and movement signatures remain partially discernible in continuous trajectories; nonetheless, local coordinate extraction provides a substantial privacy barrier compared to unconstrained RGB streaming. Consequently, we identify hybrid hierarchical architectures—wherein lightweight skeleton models handle primary real-time classification, triggering localized, low-frequency 3D video attention only when kinematic ambiguity surpasses uncertainty thresholds—as a promising direction for future research.

Edge Feasibility vs. Heavy Modern Skeleton Transformers (CTR-GCN / SkateFormer).. Recent advances in general-purpose action recognition have produced capable spatial-temporal architectures, including Channel-Wise Topology Refinement GCN (CTR-GCN) [20] and skeletal transformers like SkateFormer [21]. While achieving top-tier performance on large benchmarks such as NTU RGB+D [32], they are typically tailored for full-body 25-joint topologies with dense hand-finger and multi-segment spine keypoints. More critically, modern skeleton transformers incur large parameter footprints (often exceeding several million parameters) and heavy computational requirements, frequently requiring pre-training on massive external action databases. In contrast, automated resistance training feedback demands ultra-low latency on edge consumer hardware (e.g., smart mirrors, mobile devices, and gym-floor tablets) operating under strict thermal and battery constraints. When deployed in high-speed streaming scenarios (30 FPS, 33.3 ms real-time budget), heavy transformer models introduce significant latency and compute overhead. SkelGym addresses this operational bottleneck through custom-engineered lightweight architectures ($\approx$ 350K parameters per backbone stream, 1.91M parameters total for the full 5-stream ensemble, and only 777K parameters for SkelGym-Lite) trained entirely from scratch with zero pretraining overhead. With single-window classifier inference executing in merely 0.42–4.33 ms on host CPU (0.08–0.54 ms on CUDA, where individual backbones require 0.42–1.16 ms and the full 5-stream ensemble executes in 4.33 ms), SkelGym achieves an optimal Pareto balance between high fine-grained exercise recognition accuracy and real-time edge execution feasibility.

Limitations of Monocular Pose Estimation and Failure Modes.. While monocular pose extraction via MediaPipe Pose provides high throughput on edge processors, it incurs physical and optical failure modes that bound skeleton-based recognition accuracy. First, resistance training environments frequently introduce severe physical occlusions: barbell weight plates (e.g., standard 450 mm Olympic plates) frequently obstruct wrist, elbow, and torso tracking during deadlifts, rows, and bench presses, while spotters standing adjacent to equipment introduce visual occlusion. Second, camera field-of-view boundaries cause transient out-of-frame limb clipping, such as feet moving below frame during Romanian deadlifts or hands ascending above frame during pull-ups and overhead presses. Third, monocular 2D-to-3D landmark lifting estimates metric depth (z) relative to the hip midpoint using geometric heuristics and anthropometric priors. Consequently, movements aligned along the camera's optical axis exhibit higher longitudinal depth jitter than lateral sagittal plane perspectives. Future iterations could incorporate multi-view spatial consensus or temporal smoothing filters (e.g., learned kinematic Kalman filters) to improve landmark stability.

Dataset Class Imbalance and Long-Tail Kinetics.. A second practical limitation concerns the natural class distribution within crowdsourced fitness video datasets. Movement availability reflects gym training popularity: high-frequency compound and aesthetic movements (such as barbell biceps curls and bench presses, each with 84 videos) possess extensive diversity, whereas specialized or bodyweight stabilization exercises (such as plank with 9 videos) reside in the long tail. Although we mitigated class imbalance during optimization using Macro F1 monitoring, validation-calibrated loss weighting, and label smoothing ($\alpha = 0.05$), rare classes exhibit wider confidence intervals. Expanding the public corpus through standardized multi-angle video collection across underrepresented exercises remains an essential objective for subsequent dataset releases.

Conclusion.. In this work, we presented **SkelGym**, a lightweight, computationally efficient, and biomechanically grounded intelligent exercise recognition system for 22-class gym exercise classification. By coupling the proposed 117-dimensional Biomechanical Compound Representation with the empirically selected SkelGym-Aug configuration (unifying bilateral reflection, gravitational yaw rotation, proportional scaling, and sensor jitter) and integrating self-attention sequence modeling with 4-stream adaptive graph

convolutions via validation-calibrated soft voting, SkelGym attains **69.74% ± 1.04% window-level accuracy** and **79.11% ± 0.25% video-level consensus accuracy** (Video Macro F1: **0.7834 ± 0.0082**, Window Macro F1: 0.6882 ± 0.0068) across three independent random seeds under fixed partitions across 22 resistance exercises, with selected model comparisons verified via paired statistical tests (McNemar $p < 10^{-12}$, Wilcoxon $p < 0.005$). With an ultra-low single-window classifier inference latency of **0.42–4.33 ms on CPU** (0.08–0.54 ms on CUDA, 1.11–8.77 ms on MPS; where individual backbones require 0.42–1.16 ms and the full 5-stream ensemble executes in 4.33 ms) per 32-frame window, SkelGym enables real-time deployment on resource-constrained edge devices including smart mirrors, mobile phones, and embedded gym displays. By operating exclusively on privacy-aware 3D skeletal coordinates rather than raw RGB video streams, SkelGym eliminates the need for continuous cloud video transmission, making it suitable for privacy-sensitive home and commercial gym environments. Our comprehensive evaluation on in-the-wild multi-camera videos—rather than laboratory-controlled settings alone—demonstrates that the proposed pipeline offers a robust, practical foundation for next-generation intelligent virtual personal trainer systems.

Data, Reproducibility, and Code Availability

To facilitate rigorous scientific reproducibility and edge deployment, all project software, extracted kinematic representations, and pre-trained models are publicly hosted on Hugging Face Hub at https://huggingface.co/Cuong2004/gym-exercise-classification. The repository includes PyTorch source code for all sequence and spatial-temporal graph backbones (Transformer Mix, ST-GCN, and Four-Stream AAGCN), training and evaluation scripts, and trained model checkpoints.

The complete skeletal dataset is released in JSON, NPY, and HDF5 tensor formats, accompanied by exact temporal action segment boundaries (`.txt`) and consolidated metadata records (`Final_dataset_metadata.csv`) across Hugging Face and Kaggle. Distributing pre-extracted 3D landmark coordinates directly eliminates reliance on third-party video URLs that suffer from link-rot, while ensuring deterministic reproducibility without demanding intensive GPU pose estimation re-runs. In strict compliance with online platform Terms of Service and subject privacy considerations, raw third-party internet video recordings are withheld from redistribution. However,

the 128 author-recorded gymnasium video clips (collected with informed consent under institutional ethical oversight) and all 1,024 extracted coordinate sequences are freely accessible for academic and non-commercial research purposes.

Exact Reproducibility Protocol.. To enable deterministic replication of every result reported in this paper, the repository provides self-contained evaluation scripts that operate directly on pre-trained checkpoints without requiring model retraining:

```
git clone \
  https://huggingface.co/Cuong2004/gym-exercise-classification
pip install -r requirements.txt
# Evaluation suite across Tables 1-7:
python scripts/evaluate_local_ensemble.py
# External S&C benchmark (Table 8):
python scripts/evaluate_external_benchmark.py
# Statistical significance tests (Table 11):
python scripts/compute_statistical_tests.py
# Latency & FLOPs benchmarking (Table 12):
python scripts/benchmark_hardware_latency.py
```

All experiment configuration files (YAML configs under `configs/experiments/`), dataset metadata (`Final_dataset_metadata.csv`), partition manifests, and pre-trained PyTorch checkpoints for every backbone variant (Transformer Mix, ST-GCN, Four-Stream AAGCN Joint/Bone/Joint-Motion/Bone-Motion) are publicly hosted on Hugging Face Model Hub. This ensures 100% reproducibility of all reported tables without retraining from scratch.

CRediT Author Contributions

Xuan-Cuong Nguyen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Nhat-Quang Truong:** Data curation, Resources, Software, Validation, Writing – review & editing. **Quoc-Trinh Vo:** Supervision, Formal analysis, Validation, Writing – review & editing.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used generative AI tools for language editing assistance and code debugging support. The authors reviewed and edited all AI-generated output and take full responsibility for the content of this publication.

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Appendix A. Anatomical Landmark Mapping and Physiological Significance

Table A.15 outlines the 13 anatomical keypoints selected from Google MediaPipe Pose Heavy [8, 9]. These joints constitute the core kinematic chain required for analyzing human resistance exercises. Facial landmarks (1–10) and foot extremity points (29–32) are intentionally discarded to eliminate high-frequency tracking noise.

Table A.15: Selected 13 anatomical joints from MediaPipe Pose and their kinematic roles.

Joint ID (v)	MediaPipe Index	Anatomical Description	Kinematic Significance in Resistance Training
0	0	Nose	Coordinate origin reference anchor; head orientation tracking
1	11	Left Shoulder	Glenohumeral joint; upper-body pressing/pulling rotation center
2	12	Right Shoulder	Bilateral counterpart to left shoulder
3	13	Left Elbow	Humeroradial joint; elbow flexion/extension in curls/presses
4	14	Right Elbow	Bilateral counterpart to left elbow
5	15	Left Wrist	Radiocarpal joint; barbell/dumbbell contact point trajectory
6	16	Right Wrist	Bilateral counterpart to left wrist
7	23	Left Hip	Acetabulofemoral joint; pelvic hinge pivot in deadlifts/squats
8	24	Right Hip	Bilateral counterpart to left hip
9	25	Left Knee	Tibiofemoral joint; knee flexion/extension in squats/lunges
10	26	Right Knee	Bilateral counterpart to left knee
11	27	Left Ankle	Talocrural joint; ground contact anchor; ankle dorsiflexion
12	28	Right Ankle	Bilateral counterpart to left ankle

Appendix B. Open Science Model Hub and Checkpoint Loading

All pre-trained PyTorch [30] checkpoints, scripts, and evaluation pipelines are openly accessible on Hugging Face at https://huggingface.co/Cuong2004/gym-exercise-classification. Checkpoints can be downloaded in Python:

```
import torch
from huggingface_hub import hf_hub_download

# Download best Transformer Mix checkpoint
ckpt_path = hf_hub_download(
    repo_id="Cuong2004/gym-exercise-classification",
    filename="best_transformer_mix_aug.pt"
)
model = torch.load(ckpt_path, map_location="cpu")
print("Model loaded successfully!")
```